

Team AUTOBOTS: Bulkhead
MAE322 Fall 2025

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Bulkhead

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1. Executive Summary

Bulkhead was designed as a Search and Rescue robot (SaRR), capable of navigating various obstacles to deliver medical supplies to victims in areas where humans cannot go. Bulkhead possesses two navigational modes: an open-loop mode, controlled directly by a human operator, and an autonomous mode, in which Bulkhead can navigate complex obstacles such as ramps, narrow chutes, and walls to reach the victim. Implementing a fully integrated delivery system means Bulkhead is capable of operating with minimal user training, ideal for emergencies.

Bulkhead is an inherently simple design, featuring two large wheels as its primary means of driving and maneuvering. Bulkhead was also designed to maintain a nominal attitude during operation without assistance. Bulkhead weighs approximately 28.535 pounds and features a rear support to maintain a nominal attitude. Bulkhead has dimensions of 24 x 30 x 24 inches, enabling navigation of narrow spaces while remaining stable. Additionally, the total materials cost to build the Bulkhead was \$280. Medkit delivery was achieved by implementing a claw release mechanism, relying on gravity to provide the necessary force to bring the medkit to its intended destination. Bulkhead was built to run on an electrical system powered by a 12 V DC battery. Autonomous control is provided by a Teensy microcontroller running custom code. Infrared, light, and inertial sensors were used in concert to collect crucial environmental data for autonomous navigation.

Bulkhead's capabilities were assessed via attempting autonomous completion of a complex course on December 15, 2025. During this demonstration, Bulkhead was unable to autonomously navigate the course. Instead, Bulkhead was able to successfully operate via open-loop control, over the ramp and through the chute. In theory, Bulkhead would have been capable of delivering medical supplies as well. One additional failure was inherent in Bulkhead being incapable of clearing the wall. Completing this demonstration, Bulkhead's electrical, mechanical, and autonomous systems were shown to be fully integrated. From these results, it is clear that Bulkhead is a platform that shows great promise of becoming a fully autonomous SaRR, but that there is still work that needs to be done to reach the initial outlined objectives.

2. Introduction

The objective of designing and fabricating Bulkhead is to have a fully autonomous SaRR. This SaRR will demonstrate its capacity by navigating through a complex course resembling a realistic search and rescue situation. There are minimal limits imposed on the design of the SaRR itself, aside from being required to operate autonomously for the duration of the course. Beyond merely completing the course, Bulkhead is responsible for carrying and delivering a medical kit, weighing 1 kilogram, and being about 4 x 4 x 8 inches. Bulkhead must be able to scale a 3' wide, 8' long ramp with a max pitch of 15°. Then Bulkhead must navigate through a narrow chute, featuring doglegged segments, and approximately 3' wide as well. Bulkhead will then breach a two-stepped wall, with each step being 6 inches tall, and the first step having a depth of 6 inches.

Finally, Bulkhead will be responsible for navigating to and delivering the medkit to the victim. This process will be timed, and Bulkhead will only have one chance to prove itself, so reliability, consistency, and accuracy will be crucial goals for development.

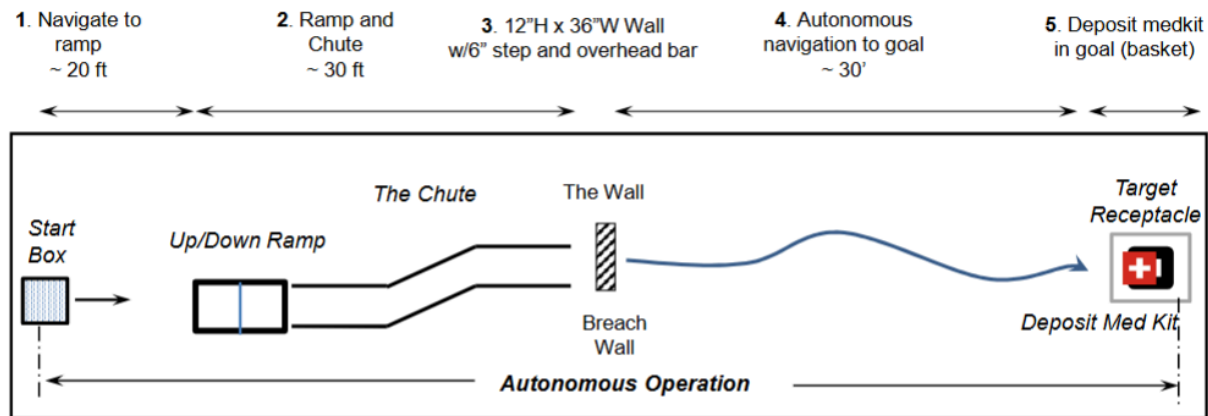


Figure 1: Notional SaRR Course

A notional version of the SaRR test course is depicted above, in Figure 1. Note that Bulkhead will start in a 3-foot box while facing away from the ramp. Bulkhead must then turn around, align itself with the ramp (which has no sidewalls), and then climb over the 8-foot-long ramp at an angle of 15° . It will then pass through the three-part doglegged chute. The chute is approximately 3 ft wide, and each segment is approximately 10 feet long. Each bend may have an angle up to 30° . After successfully reaching the end of the chute, Bulkhead must then breach a 3-foot-wide, 1-foot-tall wall. Finally, Bulkhead must locate and navigate to a target basket, or the victim, to deliver its med kit payload. This medkit is a 4-inch cubic box that weighs approximately 1 kg and has a cylindrical handle. The course navigation and medkit deposition must be conducted autonomously, although the medkit will be loaded into the SaRR manually before beginning the run.

The team drew heavy inspiration from some of the past designs used in MAE 322. In particular, the Spring 2018 robot Matilda and the Spring 2013 robot Hilux were of interest to the team. Bulkhead draws on the crucial design feature of utilizing two large wheels to negotiate and breach the wall. When initially considering design features and making plans for Bulkhead, the team agreed that breaching the wall would likely provide the greatest technical challenge. To that end, the team opted to select the simplest and most reliable design. The large wheeled design ensures that Bulkhead can easily breach by simply driving over the obstacle. It carries with it the advantage of minimizing mechanical parts, enhancing reliability, and reducing modes of failure that will need to be mitigated in the testing phase of this project.

Inspiration for the medkit design was drawn from various sources online. These designs used two claws, linked with gears to rotate in unison. Using such a system guaranteed that releases would be consistent. Additionally, the system was made simpler by using only one driving force to operate the entire mechanism, further simplifying the solution. When

considering the claw shape, it was decided to take advantage of the handle geometry. By designing the claw to completely envelop the handle, the medkit is held firmly in place, even when traversing obstacles like the wall and ramp. This differed from designs seen online, which tended to be more adaptable. However, due to the specific constraints on carrying a standardized medkit, it was possible to optimize the claw design to maintain security and simplicity.

The overall design philosophy of Bulkhead was to be visually appealing to customers, durable, handable, and simple. In a real disaster area, the durability of the robot is critical in unexpected environments. Components of the design cannot be easily exposed to potential damage. Furthermore, simplicity allows for rapid scaling and deployment - a feature critical to saving lives. Finally, in the hopes of creating a commercially successful product, Bulkhead's appearance is more important than one may think. The SARR must appear functional, well-designed, easy-to-use, and welcoming to those working in stressful disaster zones. The basic design concept of Bulkhead is shown below in Figure 2. Bulkhead consists of a rectangular chassis with two 24" wheels (left and right) and a rear wheel on a back plate that extends out the back of the chassis. The medical kit and all electrical and mechanical parts are stored inside the chassis, and a servo-powered claw on the top plate of the chassis secures the handle of the medical kit until Bulkhead reaches the basket. The drive wheels are powered by one VEX Mini-CIM motor each, geared with a 100:1 ratio followed by a 3.3:1 belt and pulley. On the left and right sides of the chassis, a distance sensor is mounted on a plate that extends the sensor out in front of the wheels. On the underside of the chassis, there are also a pair of photoresistor light sensors mounted low to the ground that Bulkhead uses during its light-finding mode in autonomous navigation.

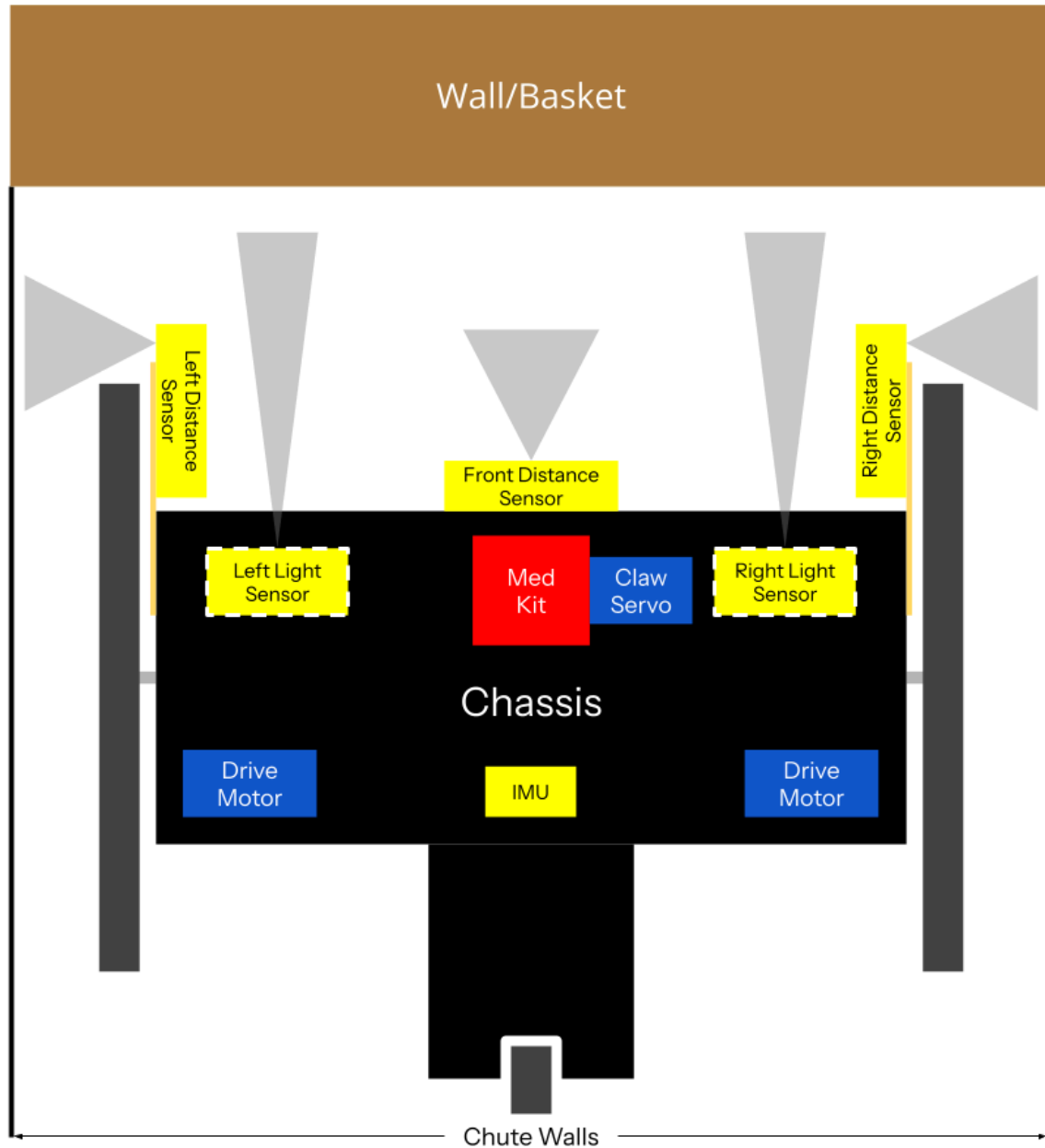


Figure 2: Bulkhead Simple Schematic Drawing

Each design pillar will be discussed in the following section. For each element, the advantages of pursuing the objective, as well as how the objective is being achieved, will be considered. Finally, the ultimate success in pursuing the goal will be assessed. Starting with weight consideration, keeping the SaRR light will make it easy to deliver the SaRR to the location of the victim. Additionally, being light will make getting over obstacles such as ramps and walls less energy-intensive, increasing the operable range. Designing Bulkhead to be robust will improve field operations. Being able to take hits, falls, drops, and rough transport will ensure Bulkhead can be consistently deployed and operate in harsh environments without failure.

Robustness was tested to some extent during the demonstration, as a drop test showcased the SaRR's resilience. Further testing was not really possible, but this shows promise. Additionally, Bulkhead remained relatively simple. The mechanical and electrical systems were both designed to be easily replicable. The parts used were standardized, with duplicate elements being used in order to simplify the number of unique parts that needed to be produced. The number of motors and sensors was reduced as much as possible to limit power consumption and to simplify calculations. All information was run through the Teensy microcontroller, greatly simplifying the operations. The end result was a fairly simple electrical system, detailed in later sections. Finally, size considerations were considered, but not optimized. The team recognized that the SaRR must be capable of completing the course, but did not see a pressing need to minimize the size beyond that.

Looking beyond the prototyping phase, it will be crucial to make Bulkhead more robust and reduce complexity without sacrificing operational capabilities. Improvements to the drivetrain reliability, of which the team faced significant issues, are also essential. This will ensure that Bulkhead can complete more operations before needing servicing, be easily repaired in the field, and cost less to purchase and manufacture. Finally, if Bulkhead is to be utilized in areas where widespread disaster has struck, it will be crucial for the SaRR design to be constructable using minimal resources. To that end, the team has opted to use scrap materials to simplify manufacturing and minimize the cost of construction. While there were still significant expenditures, these were primarily on elements that could not be constructed from scrap materials or were used for aesthetic purposes. Nearly all of the critical, load-bearing elements of the design were created from scrap, validating our design goal of creating a SaRR that can be built in devastated regions.

Finally, in the course of development, the team took steps to ensure that Bulkhead operates in a safe manner. The team considered adhering to engineering standards, so as to speed up Bulkhead's adoption in international search and rescue operations. However, concerns around time and development led to the abandonment of this concept. Additionally, the team primarily envisioned Bulkhead being constructed and deployed in devastated regions, where these standards would be of relatively minimal importance beyond being a potential target for the design to pursue. The team would be open to considering adherence to these standards in the future, but felt it exceeded the possible goals of the course.

3. Specifications

3.1. Weight and Size

Weight	28.535 lb
Length	24"
Width	30"
Height	24"

3.2. Target Time to Deliver Medical Kit to Victim

Prototype tests of Bulkhead showed that the maximum speed is calculated to be 44 ft/min in the next section, but the nominal speed is 1.047 ft/s. During autonomous navigation, it is advantageous for Bulkhead to travel more slowly to increase the reliability of the autonomous mode. Note that there is a discrepancy between the measured speed and the calculated maximum speed. This likely means calculations were off, with internal differences being unaccounted for.

3.3. Estimated Maximum Speed and Range

After accounting for the 100:1 gearbox, the true rotational speed of the wheels is $1400 \text{ rpm} \cdot \frac{1}{100} = 14 \text{ rpm}$. Then, multiplying by the circumference of the wheel to obtain the linear speed $14 \text{ rpm} \cdot \frac{24\pi \text{ in}}{1 \text{ rev}} = 1055.6 \text{ in/min}$. This is quite fast, and Bulkhead should move at a slower speed, which lends itself to consistent motion because there is less slipping when navigating autonomously. This slower speed can be chosen arbitrarily, so it was decided to use 50% of the maximum speed. Therefore, Bulkhead's maximum functional speed is $5278 \text{ in/min} \cdot 0.25 \cdot \frac{1 \text{ ft}}{12 \text{ in}} = 44 \text{ ft/min}$. Notionally, this also allows for Bulkhead to speed up at any time in case the notional victim needs faster assistance.

To move Bulkhead at the slower 50% speed, the motors will rotate at $1400 \text{ rpm} \cdot 0.50 = 700 \text{ rpm}$. This speed corresponds to 8.3A of current draw. With two motors, they will draw 16.6A combined, and at 12V, the power consumption is $16.6 \text{ A} \cdot 12 \text{ V} = 199.2 \text{ W}$. The battery that we intend to use with Bulkhead is a 12V, 10Ah Dakota Lithium battery. This battery has a capacity of $12 \text{ V} \cdot 10 \text{ Ah} = 120 \text{ Wh}$ (watt-hours). To calculate the battery lifespan of Bulkhead, we divide 120 Wh by 199.2W to get 0.6 hours or 36 minutes. Given Bulkhead's calculated speed, this gives it a range of $36 \text{ min} \cdot 44 \text{ ft/min} = 1584 \text{ ft} = 0.3 \text{ miles}$. Notably, this is significantly longer than the demonstration course Bulkhead is expected to complete.

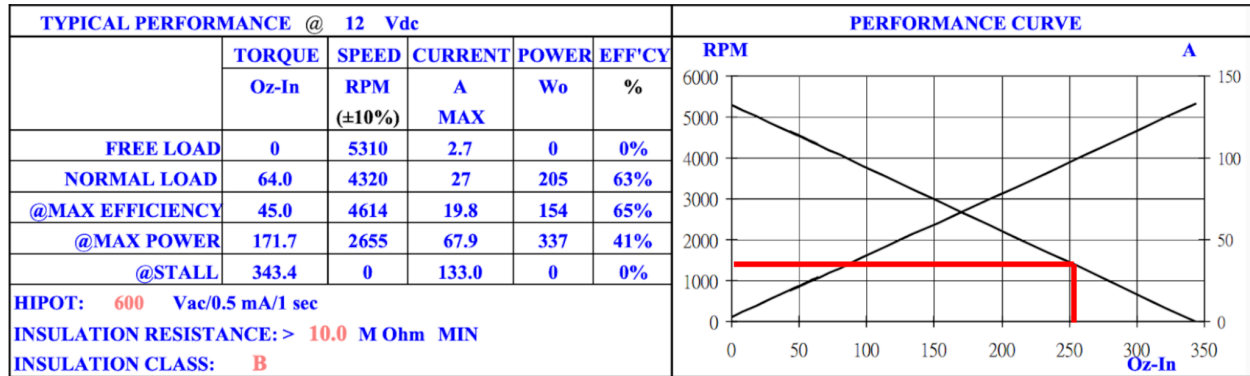


Figure 3: Mini-Cim Motor Chart

3.4. Operational & Navigational Modes

Bulkhead contains two operational modes, depending on the state of the switch on Channel 5 of the controller. In open-loop mode, the controller's right joystick controls the motion of the robot using "arcade control," where pushing the joystick up/down/left/right corresponds with moving the robot forward/backward/left/right. In closed-loop mode, the robot moves autonomously by progressing through 8 navigational modes designed to enable Bulkhead to complete each section of the course. These navigation states are:

- State 0: Navigating to the light on the ramp
- State 1: Ascending the ramp
- State 2: Moving over the top of the ramp
- State 3: Descending the ramp
- State 4: Navigating the chute
- State 5: Breaching the wall
- State 6: Navigating to the light on the basket
- State 7: Dropping the medkit into the basket

The autonomous code switches states based on sensor readings from an inertial measurement unit (IMU) and front distance sensors. State 0 describes the robot's navigation towards a light on the ramp using photoresistor sensors. The state switches to State 1 when the IMU detects a tilt while ascending the ramp corresponding to a value larger than a +3 x-acceleration value. The state switches to State 2 when the IMU senses a return to horizontal position at the top of the ramp, or when the x-acceleration value is greater than -0.6 and less than 0.6. State 3 begins when the IMU detects a downward tilt while Bulkhead descends the ramp, corresponding to an x-acceleration value of less than -3. Once the IMU detects a second return to horizontal position, the robot is understood to have cleared the ramp and switches to State 4, navigating the chute. Bulkhead senses the completion of chute traversal when the front distance sensor detects the presence of the wall in front of it, shifting to State 5, which causes Bulkhead to

continue driving forward until the IMU detects another return to a steady horizontal position based on z-acceleration values. Since the tilt that Bulkhead experiences during wall traversal is much larger than during ramp traversal, z-acceleration provides a better specification of values. Once the wall is cleared, Bulkhead moves into State 6 and initiates another light-following mode. When the photoresistors detect the light on the basket, Bulkhead drives forward until the front distance sensor detects the basket and stops movement, entering State 7. In this final state, Bulkhead releases the Medkit into the basket using a servo motor connected to the claw holding the Medkit.

For initial testing, Bulkhead was operated manually to check the system's hardware and software functionality at different parts of the course. However, the ultimate goal is to traverse the entire course autonomously, beginning from an initial drop test oriented away from the ramp, and ending with the Medkit delivery.

4. Project Management

4.1. List of Team Members and Responsibilities

- Arthur Balva - CAD lead, design & CAD of chassis, medkit dropper, wheel mounts
- Dwayne Bazil - manufacturing, motor mount CAD
- Connor Brown - manufacturing, wheel hub design, back platform design
- Jack Byrnes - overall quality assurance, assistant to all subgroups, coordination of reports and presentations
- Evan Callas - manufacturing lead, electrical integration, CAD & assembly assistance
- Lucas Denault - back platform design, manufacturing & assembly
- Tyler Harris - CAD lead; medkit dropper and rear plate design; FEA; manufacturing
- Ani Kozak - manufacturing & assembly, robot background research
- William Li - autonomous programming, electrical integration, assistance with CAD
- Sophia Verwoert - autonomous programming, electrical integration, assembly
- Sofia Zypman - autonomous programming, manufacturing

4.2. Key Tasks

Task	Team Member(s)
Drivetrain Design	Arthur Balva
Rear Plate Design	Connor Brown, Evan Callas, Lucas Denault, Tyler Harris, Ani Kozak
Electrical Integration	William Li, Sophia Verwoert
Mechanical Integration	Arthur Balva, Dwayne Bazil, Jack Byrnes, Evan Callas
Autonomous Programming	William Li, Sophia Verwoert, Sofia Zypman
Medkit Delivery System Design	Arthur Balva, Tyler Harris, Jack Byrnes
Manufacture of Parts	Jack Byrnes, Tyler Harris, Connor Brown, Lucas Denault, Evan Callas, Sofia Zypman, Ani Kozak

4.3. Schedule:

When defining the milestones for the team to meet, the mandatory deadlines imposed by the course itself were considered. The milestones defined by the team were set to be optimistic, with issues expected to arise over the course of designing, manufacturing, integrating, and testing Bulkhead. To remain on track, the team lead ensured that subteams were capable of completing their tasks, while also reassigning team members to provide support on the most pressing issues

or time-consuming tasks. It was anticipated that some events would move around as the timeline progresses, and the schedule was made with this in mind. The tables below compare the initial milestones set by the team with those defined by the MAE 322 Administrators.

Week Of	Objective	Subteam
29 September	Demonstrate Drive Train (Open Loop)	All Team Members
20 October	Preliminary Design Presentation	All Team Members
24 October	Preliminary Design Report Due	All Team Members
27 October	Autonomous Navigation to Light Source	Autonomous Subteam
3 November	Ramp and Chute Autonomous Navigation	Autonomous Subteam
17 November	Traverse Wall (Open Loop)	Mechanical and Electrical Subteams
1 December	Autonomous Medkit Delivery Demo	All Team Members
8-10 December	Full Course Rehearsals	All Team Members
12 December	SaRR Completely Finished	All Team Members
15 December	Final Presentation and Demonstration on the Course	All Team Members
15 December	Final Design Report Due	All Team Members

Figure 4: Key Milestones as Defined by MAE 322 Administrators

Estimated Date	Objective	Subteam
24 October	PDR Completed	All Team Members
31 October	Preliminary CAD Completed	Mechanical Subteam
7 November	Chassis Manufacture Complete	Mechanical Subteam
7 November	Wire Analysis Complete	Electrical Subteam
7 November	Autonomous Navigation To Light and Breach Wall	Autonomous Subteam
14 November	Bulkhead (SaRR) Assembled	Mechanical and Electrical Subteams
14 November	Test Bulkhead - Open Loop Wall Breach	Mechanical and Electrical Subteams
21 November	Finalize CAD Model	Mechanical Subteam
21 November	Make and Finalize Tweaks to Bulkhead Post Test	All Team Members
21 November	Successfully Complete the Whole Course, Assess Multiple Times for Repeatability	All Team Members
5 December	Complete Final Design Report	All Team Members
12 December	All Final Touches Complete, Bulkhead Fully Ready To	All Team Members

	Complete SaRR Course	
15 December	Demonstrate Bulkhead Success, Give Final Design Presentation	All Team Members

Figure 5: Nominal Initial Milestones Set Out By the Team

The final schedule that the team followed ended up failing to follow these goals. This was mostly due to the various issues that cropped up during the course of making Bulkhead, and are detailed in Section 4.4. A Gantt chart of some of the various tasks is listed below. Note that the various subtasks often ran past their intended deadline. This was due to both the risks as stated below, and due to underestimations of the time and amount of work associated with each milestone. Additionally, several of the key milestones required the resolution of multiple problems before they could be completed.

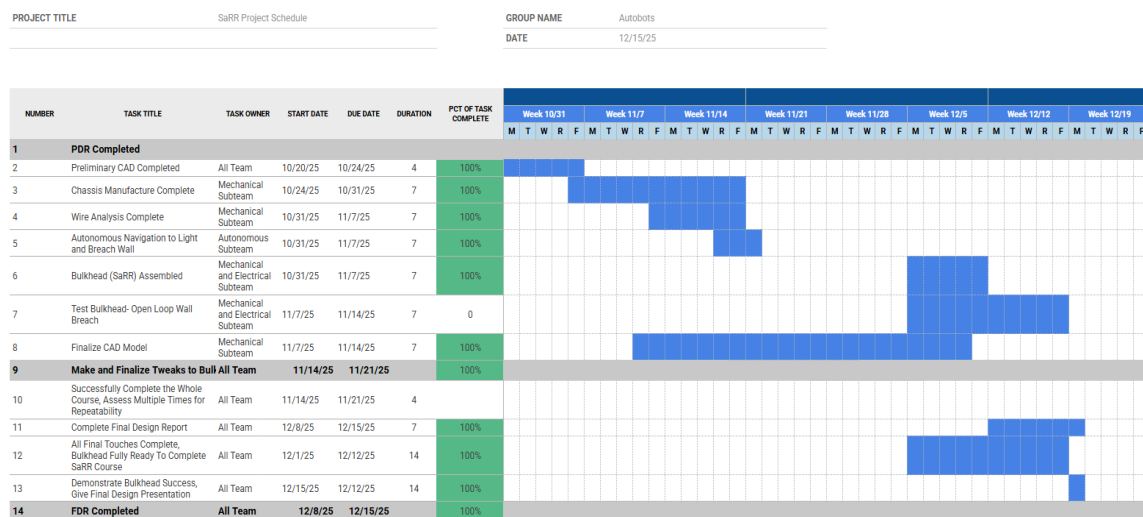


Figure 6: Gantt Chart of Tasks for Bulkhead

4.3.1. Hours Estimation and Comparison:

In the table below, the team has gone ahead and made two side-by-side estimations. While working to complete the project, it was necessary to estimate the total time required to complete various tasks, as well as the project itself overall. As the table below shows, the estimations made by the team were very far off from the mark. These estimations were made with the knowledge that projects often take significantly longer than anticipated to complete. The lack of accuracy, despite this knowledge, really drives home the importance of building in

excessive amounts of time to complete tasks, as well as prioritizing remaining on schedule, or nearly on schedule, to complete final goals.

Task	Estimated Time (hours)	Actual Time (hours)
CAD Design	50	100
Initial Manufacturing	40	190
Initial Electrical Work	60	30
Initial Autonomous	20	30
Troubleshooting/Problem Solving	80	200
Total	250	550

4.4. Risks:

4.4.1. Identified Risks

Scheduling

One identifiable risk we ran into was the front-loading of our timeline. While this was initially beneficial, we ran into issues when the computers in the machine shop were down for just over a week, significantly delaying progress when we had already run into the issues listed above. Unfortunately, the only way to deal with this issue was to delay our entire timeline by a week and allocate many more hours than we had originally planned to. The fact that we had attempted to front-load the timeline of completion but still ran into issues is evidence that you really must overload the front of your schedule to a large degree so you have ample time to iterate and troubleshoot problematic areas before testing.

Autonomous Navigation

Another risk we identified before beginning the construction of Bulkhead revolved around programming concerns. True testing of the autonomous system is essentially impossible without a finalized/working mechanical design. In our case, we were dealing with massive mechanical difficulties until the last minute, largely stemming from some of our scheduling setbacks, because we could only test the autonomous code on the mock chassis; we had no real understanding of whether the code would work or not.

4.4.2. Unidentified

However, not all risks were identifiable and came up without warning. Some of the initial risks were design issues and timeline issues. For the mechanical risks, most of our initial design of the robot depended on a variety of mathematical models in which we specifically included various safety factors in an effort to ensure that, although our modelling might have been rough, components would still work as expected. As we completed the assembly of Bulkhead, we realized that some of our assumptions had proven to be incorrect, specifically pertaining to our drive shaft sub-system. Based on our initial design, we had concluded that in order to best optimize weight while maintaining proper power output, we would use PLA to construct the wheel and axle mounts. We quickly found this risk to be too great and went through a variety of different designs before eventually landing on one that would not shear when a greater torque was applied. This issue was mostly material in nature, as we had overestimated some of the material properties of PLA and had to switch to aluminum scrap from the shop. On top of these assumed risks, there were a couple of unforeseen issues that arose pertaining to the issue with the computers. Once we had addressed the fact that the computers were down and understood that we'd then need to put more time into the robot the next week, we attempted to schedule to have a TA in the shop and had difficulties with people not being present.

Another unidentified risk we ran into was the timing belt and the power distribution from the motor to the wheels. Specifically, we had assumed that by bolting the motor mounts at a set location, we would be able to provide adequate tension in the belt without slipping. As we were only able to test on the last day of the project, it was realized that the belts would slip when attempting to get over the wall, so two tensioners were fashioned rather quickly to remove the slipping, but at the end of the day, time ran out. As discussed above, on top of the slipping, many iterations of the wheel hubs were necessary to prevent failure, a problem resolved so late that there was very little time left to test both the autonomy and the slipping of the belt.

4.5. Overall Project Management

4.5.1. Team/Subteam Organization

To ensure the successful completion of the SaRR project, the team has opted to split into several subteams. These subteams are as follows: electrical, autonomous, mechanical design (CAD), and mechanical manufacturing. Different objectives drive each of these teams, and all fall under the supervision of an overall team leader. The objective behind creating up to 4 separate teams is to allow parallel workflows to be created. Establishing these parallel workflows allowed the team to work simultaneously on different elements of the SaRR. This did make it easier for issues to be mitigated as they arose. However, despite attempting to avoid it, issues still arose that endangered the completion of Bulkhead. The roles and leadership information for the various subteams are expounded upon in the following paragraphs.

The Mechanical subteam was split into two parts: CAD and manufacturing. The subteam leads for the CAD section of the SaRR were responsible for assigning CAD tasks to other members and keeping the design documents organized for ease of adjustment and manufacturing. The CAD team was responsible for designing the chassis, drivetrain subsystem, medkit delivery,

and wall traversal mechanism, as well as conducting mechanical analyses on crucial elements of the mechanical design of Bulkhead. The manufacturing subteam was responsible for directing the manufacturing schedule by delegating jobs to ensure completion on time, and reviewing that CAD designs are consistent with the team's philosophy of a simple and easy manufacturing process. The manufacturing subteam was also responsible for overseeing the assembly and integration of the various subsystems.

The objectives of the electrical subteam involved the integration of the three motor controllers, necessary sensors, the acquisition and organization of physical parts required for software implementation, reviewing power calculations to ensure correct output, and the design of an accessible circuit board. Part of these responsibilities included the manufacture, assembly, and organization of various wires to ensure adequate power delivery.

The Software team developed and tested code responsible for the successful navigation through the chute, light detection, and the successful autonomous completion of the course using sensor readings. To ensure the completion of this objective, it was necessary to ensure smooth integration between mechanical components and sensors with the software. It was also crucial to alter and fine-tune the code of the SaRR during the testing phase to iron out bugs and improve the reliability of Bulkhead.

In the design phases of the project, the subteams worked on their separate tasks, but as the project progressed, an emphasis was placed on being flexible. To compensate for the fact that everyone has other classes and responsibilities, keeping track of the team's progress and being willing to step into other subteams became essential for the sake of completing the SaRR. During the implementation and testing phase, reworks to the design concepts needed to be made on a moment's notice, so clear communication and logs of various iterations allowed for ease of adjustment.

4.5.2. Intrateam Meeting and Communications

To keep the team aligned with required checkpoints and make the most of pre-allocated time, work was mainly completed during the Tuesday lab section. During these sections, the team would gather to discuss progress and targets for the week. To brief members of the team who were unable to make lab meetings, phone calls, and in-person meetings with the team lead were scheduled to keep all members updated and involved with the team. Manufacturing tasks and leftover goals were assigned to the Thursday lab sections as necessary. Finally, additional time was reserved on Friday afternoons for completing crucial tasks that needed urgent completion. Subteam leads organized separate times to meet their members and kept track of their progress, allowing each team to progress without the constraints of finding time outside of the lab to meet as an entire group. All online communication was done via a group chat with all members, and files containing design concepts and updates were kept in a shared Google Drive. Additionally, members were encouraged and often reached out to the team lead regarding conflict, requesting information, and seeking guidance. CAD files were in the shared MAE network drive for ease of access. Post Thanksgiving break, meetings were scheduled by

availability, with the Team Lead organizing these work sessions. This strategy of flexibility was crucial in a non-standard working environment. Overall, intrateam communication was generally successful, with members completing tasks when possible, without being at the expense of other courses.

4.5.3. Issue Resolution and Responsibility Strategy

To ensure the SaRR project is completed on time, the Team Lead had the responsibility to organize the team, ensure clear communication between subteams, and keep progress in line with the timeline. As part of this, it was necessary to oversee all elements of the integration and final assembly. Another key responsibility was to ensure that the PDR and FDR reports and presentations were completed on time and accurately represented the ideas and processes of the team. As Lead, they also hold the responsibility of regulating interpersonal communication. While no major issues came up, keeping track of member attendance and ensuring communication was open between team members was integral to getting work done. The Subteam leads had the responsibility of briefing the Team Lead as well as other members of the team on the progress being made.

Despite being organized into subteams, all team members were expected to help and support other teams where needed, by assisting with CAD designs, stepping in when another member of the team had other engagements, and keeping open lines of communication regarding all aspects of the SaRR. This emphasis on support and collaboration should reduce the individual workload of this project and ensure the team can successfully demonstrate Bulkhead in December. Giving ownership of different tasks to every team member allowed people to feel responsible, proud of the work that they contributed, and hold themselves accountable for their own subtasks.

4.5.4. Project Management Lessons Learned

During the course of completing this project, the intricacies of project management and team leadership were made exceedingly clear. Some places where the management was successful were in keeping morale up, keeping goals defined, and keeping team members motivated to complete the necessary work. This was done primarily via maintaining involvement, projecting optimism, and allowing for jokes and group dynamics to develop. In particular, meeting structures were loose and informal, allowing people to speak freely. This allowed for the development of interpersonal connections. These connections then allowed for greater collaboration to occur during the manufacturing and troubleshooting stages of the project.

Despite these successes, there are a number of things that I would change in hindsight. There are two major things that would have allowed for better management of the team. These two changes that should be made are the clarification and publication of the necessary tasks, and greater emphasis on completing objectives on time. Firstly, the publication of necessary tasks would have made it exponentially easier to communicate with team members. Instead of a central location containing all of the information needed to construct the SaRR, team members

would reach out to the team lead for direction. At times, the team lead would also be lost on what tasks were assigned to whom, leading to work inefficiencies by assigning multiple people to a single task. Using some online websites, it would be possible for all tasks to be organized by status, visible by every member at all times, and clear up some of this confusion. Additionally, in the future, the team lead should be more forceful about the completion of tasks and objectives. While there are obviously reasonable causes for delaying the completion of tasks, the team let these tasks pile up. Unfortunately, it was decided that it would be better to keep team morale up, at the cost of slipping deadlines and completion dates. This was mostly due to the lack of an established hierarchy, a desire to keep the team working together, and the non-traditional structure associated with working on a group project that was academic in nature. Despite these hurdles, the team lead would like to focus on implementing new leadership strategies that are able to obtain results while still promoting positive social interactions.

5. Design and Analysis

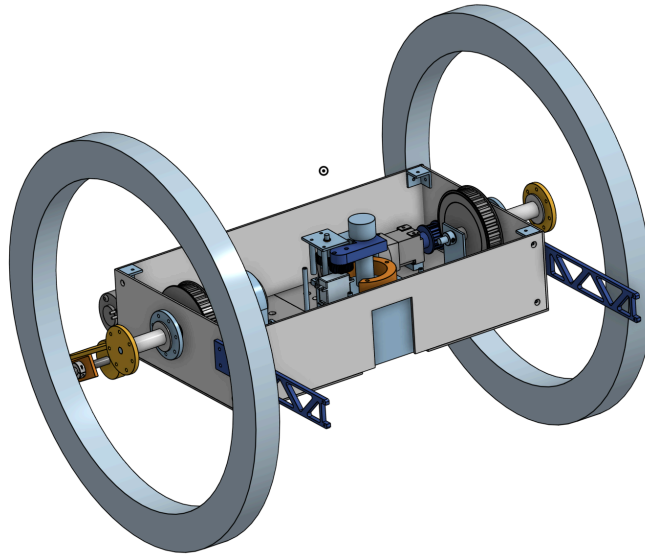


Figure 7: Overall CAD of Bulkhead

5.1. Weights and Center of Mass

The team worked to minimize the weight of the SaRR- the final weight was measured to be 28.535 lbs, mostly concentrated at the bottom of the chassis. Subsystems: While the chassis was responsible for a large majority of the weight, we did find the back plate itself (without the rear motor mounted) to be about 1.75 lbs. This leaves the remainder of the weight distributed amongst the main chassis and wheels. Drawings of key members can be found in the Appendix.

The CoMs of the SaRR and its subsystems were manipulated with stability and energy efficiency in mind. Despite these considerations, given Bulkhead's bottom-heavy chassis, the center of mass was less important than rotational stability, which the SaRR has achieved.

As mentioned above, the weight of the rear platform was roughly 1.75 lbs. For ease of understanding and due to the symmetric nature of the rear platform, we found the CoM to be roughly around the geometric center of the plate. However, because of the original rear-wheel drive design of the place, there is a large chunk of acrylic taken out on the left-hand side that would have made room for another belt to drive the rear axle, so we shifted the CoM slightly to the right side of the plate.

For the CoM of the main chassis, we designed this sub-system to have a low CoM that can be found in the back portion of the chassis. This design choice was made so that we could reduce instability and forward rotation while driving. We found that while Bulkhead was driving,

we had no issues with rotational instability like forward rocking, indicating that our CoM was far back enough for the goals of this project.

Finally, we decided to find the actual CoM and, given the fact that the rear platform only made up about 6% of the total mass of our S.a.R.R., we felt it was a reasonable approximation to take the total CoM to be slightly below and behind the main chassis CoM. Again, this approximation is valid, as we could see during driving and testing, our robot had no issues with forward rotation while driving forward or turning. Although looking into the harsh conditions undergone during the wall traversal, it would have been ideal to have an even lower and further back CoM to ensure no forward rotation when our angle of attack was so steep.

5.2. Design Loads

5.2.1. Horizontal Configuration of Load-Bearing Members

5.2.1.1. Wheel Axles

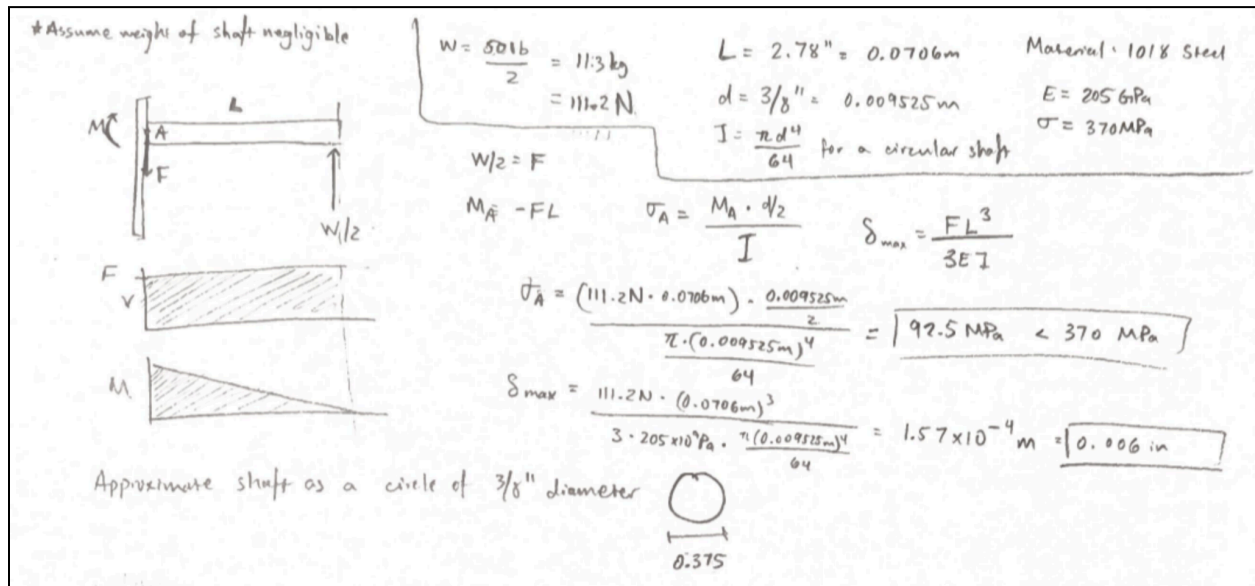


Figure 8: Drivetrain Axle Normal Operations Cantilevered Beam Calculations

Figure 8 above shows calculations for the stress and deflection experienced by the drivetrain shafts, which are supported at two locations inside the chassis but then cantilevered on the outside of the chassis. The wheel axles carry the main load of the chassis, with the only secondary support from the rear platform. According to the calculations above, the $3/8"$ circular shaft made of 1018 steel will only deflect by 0.006 during normal driving operations (a negligible amount), and the stress experienced at the mounting point is less than the yield stress of the material.

The second consideration with the cantilevered axles was bending during the drop test, and when Bulkhead breaches the wall. The main consideration for the drop test was any plastic deformation that may occur when Bulkhead impacts the ground. Considering the full axle system (not just the cantilevered portion), to prevent stress from concentrating in any one location, we created two reinforced axle mounts that were connected to the main chassis. Within the axle mounts, the shaft passed through a plain shaft bearing. With this design, the load is distributed over two contact points and over a larger surface area in the chassis. Returning to the cantilevered axle calculations, shown in Figure 9 below, after Bulkhead breaches the wall or during the drop test, it falls a distance of 12 inches. Using conservation of energy, the potential energy is converted into kinetic energy, and the velocity at impact with the ground is calculated. Then, using impulse and assuming that the bike tires spread out the impact over a time of 0.25 seconds, we find that the force imparted to the shafts during a drop is actually very similar to the force applied during normal driving operations, which was already proven not to pose a problem for the bending of the axles.

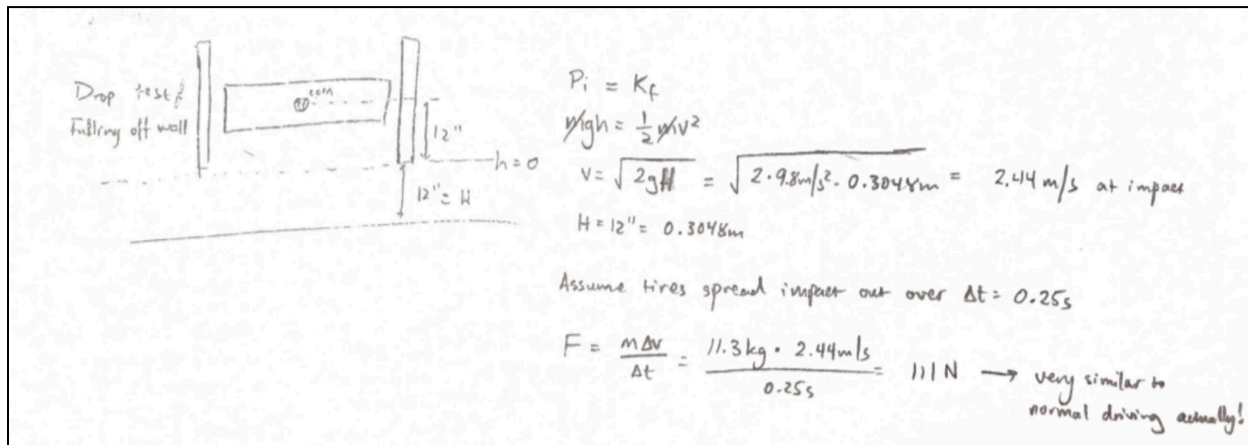


Figure 9: Force on the Cantilevered Shaft from Drop Test

One thing to mention is that the battery within the SARR was off-center, but that displacement of the center of mass to the right side is negligible for the load analysis. Because that is a larger mass that is off the center line, it applies torque on the SARR, but the sturdiness of the chassis was able to withstand this negligible torque. Another noteworthy torque was the medkit at the front of the SARR, which, when thinking about the load it applied, was not as important as the center of gravity. What it truly did was bring the center of mass forward in the chassis, so the analysis will just be on the center of mass that takes this into account.

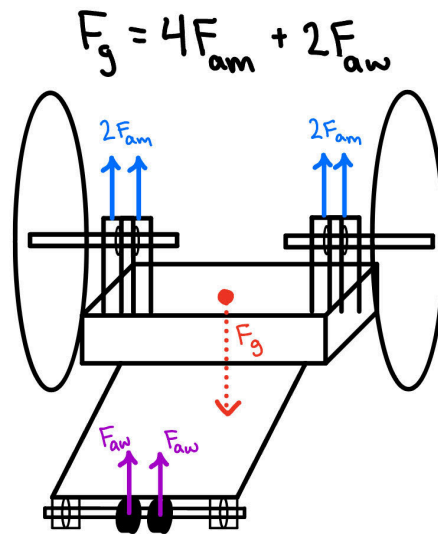


Figure 10: Vertical Force Analysis in Assembly

Here we see that the gravitational force that acts on the center of mass is being directly opposed by the force applied by the four axle mounts (supported by plate attachments) and the two wheels on the rear platform. We can see that these are just normal forces and can easily oppose the F_g .

5.2.1.2. Up the Ramp

Pointless to do a torque analysis on the horizontal forces; this creates a bending moment, but our chassis is firm and stable enough that these torques are negligible. Essentially, any analysis here would be completely redundant and unnecessary. Therefore, we do not seek findings for concision.

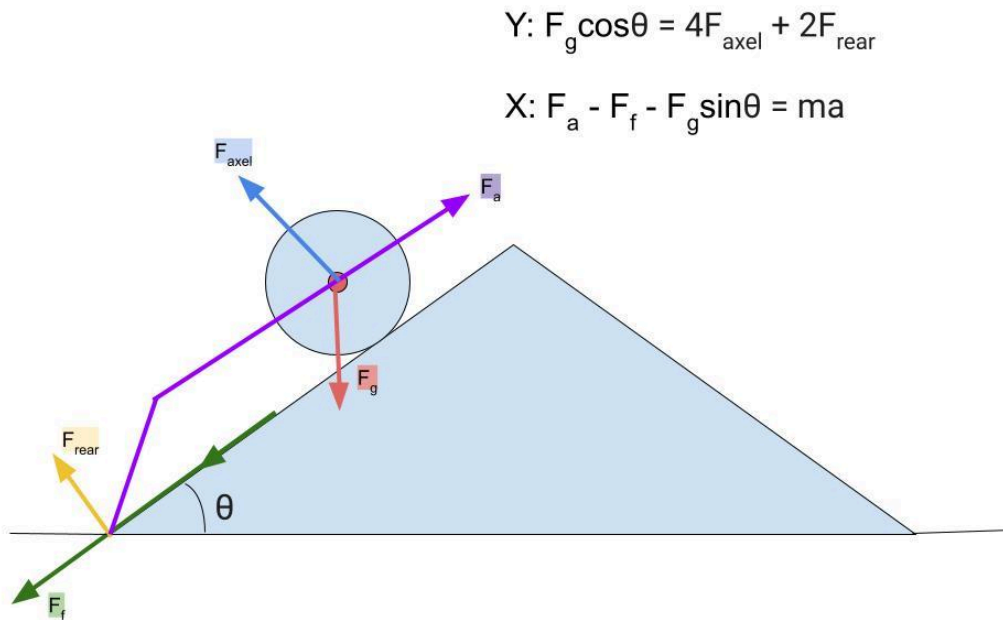


Figure 11: SARR force diagram up the ramp

Here, there is now a force that we need to assess: the acceleration. Here, the force applied can overcome the friction and gravity because the friction is very small, but the SARR is heavy enough that it is not likely to slip.

5.2.1.3. Down the Ramp

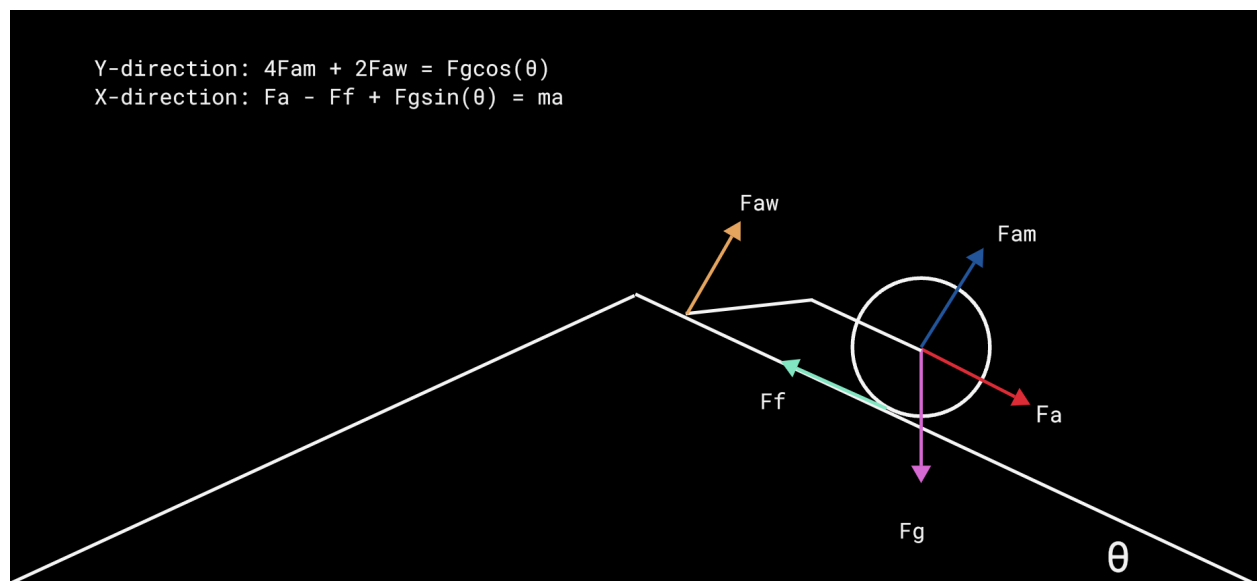


Figure 12: SARR force diagram down the ramp

Here, we can see that the main difference is that gravity is now with the force applied from the belt and is now being ADDED to that of gravity. This was not initially thought of, and made it so that on the declining portion of the ramp, Bulkhead experiences a rapid acceleration and can cause difficulty to control for our autonomous code going into the ramp section of the course. This rapid acceleration was mitigated by letting go of the throttle after peaking on the ramp.

5.2.1.4. Rear Platform During High Impact

The final phase to analyze was the big tires touching the ground and the rear platform following it after the wall sequence. We were scared that the impulse of the ground to the rear platform was going to be too large, and shear the plastic. We were attempting to measure this value with FEA, but without proper material properties and bouncing mechanics, we decided to test qualitatively. Also, the bending moment locations were different with and without the motor, and we needed to see how high we could drop it from before it broke. The accepted shear stress [A] was incredibly high for the cast acrylic, and we wanted to test that for ourselves. We grabbed a similarly sized piece of acrylic (4" x 14" x 1/4"), fixed it around a central rotating rod, and dropped it from a few different heights. Without the motor, everything seemed fine. However, once we dropped it from parallel with the motor attached at the center, it fractured immediately. So, our solution was to add 2 aluminum 6061 T brackets (scrapped from the shop) on the bottom surface of the rear platform, which gave more support in the direction of bending. This straightened the platform out, and now, even with the motor on, we could drop the acrylic from the 1-foot high point and it would survive.

Stage II:

$$Y: T_g = F_g * r$$

Instantaneous angle: θ

O is fixed rotation, P is point of contact

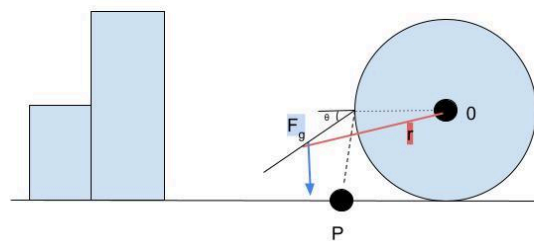


Figure 13: Rear Platform Falling to Ground After Wall

There were some mechanics that we needed to consider based on the wheels for the back platform. The rubber gave some shock resistance and a bouncing effect, which had a greater effect than the acrylic on the ground during the initial testing. This had 2 consequences, first that

not all the gravitational energy went straight into the acrylic, that bouncing back eased the transition. Second, it created more vibration and dissonance in the system at high-impact moments. This required us to velcro the battery and electronics down to mitigate the post-impact up and down movement.

5.2.1.5. Bottom plate near the medkit opening

This plate had a strict constraint on it, it needed to be above 9” to ensure that we could drive over the box. This was a crucial step in our methodology for Bulkhead, and was a strict constraint throughout the design process. We initially were scared because when putting all the electronics in the chassis, there was a sagging effect on the front of our bottom plate near the medkit, dropping it by about .5”, causing a run-in with the basket. However, after putting in the standoffs and fully attaching the top enclosing plate, there was no sagging, unity with the dimensions from the CAD model.

5.2.1.6. Claw arm

To consider how the drop test would affect the rear platform, it was slightly lifted off the ground and dropped to observe where along the acrylic it flexed the most. The upper part of the acrylic closest to the main chassis showed the most flex, so to reinforce that area, metal plates were added to both sides of the acrylic. These plates were added to connect existing metal braces that ran perpendicular along the acrylic, effectively distributing the impact along the plates instead of focusing on one spot. These braces will also support the SaRR through the conclusion of its wall traversal by absorbing the impulse from the rear platform swinging down to the ground once the main wheels settle.

Throughout the course, the claw was designed to maintain the Medkit in a fixed position relative to the chassis by using a normal force that counteracted the weight of the Medkit and a friction force between the Medkit and the claw’s surface to restrict any motion that the Medkit could experience while Bulkhead was in motion. During ramp navigation and wall traversal, Bulkhead’s rapid acceleration and deceleration also increased the effective load acting on the claw through added inertial forces at the Medkit-claw interface. Finally, during drop-off, a servo motor actuated the claw, releasing its grip. The supporting normal force became zero, allowing the Medkit to fall into the basket.

5.3. Design Changes

5.3.1. Rear plate/arm

One of the biggest design changes that we made between the PDR and final design of Bulkhead revolved around the rear arm. Initially, we had designed the rear arm to simply be for stabilization, a singular arm centered on the chassis that would extend and have a single, rolling wheel. However, when we began to design this part, it quickly became apparent that if we

wanted the arm to extend behind the robot as well as down at some angle, this single wheel design could prove problematic. We were worried that if there were any major perturbations Bulkhead might be rocked off balance and be unable to continue moving. We decided to then extend the back arm into a plate with a larger width and incorporate two rear wheels placed slightly apart for balance. Around this time we became concerned with Bulkhead's ability to successfully navigate over the wall, after realizing that some of our mathematical assumptions might not hold as true as we thought. So we designed the back plate to hold another motor so that the rear axle would be driven to provide some extra torque to traverse the wall. While this solution seemed valid at first, we quickly realized that the presence of a third motor drastically increased the overall weight of Bulkhead, and we determined that the added torque from the rear wheels would not overcome the added weight from that motor. For a while we did mess around with the idea of attaching a smaller motor to at least provide some driving mechanism. We concluded that it would be easier for the back plate's sole purpose to be balance-related. This is the final design that we landed on and implemented.

5.3.2. Claw System

Another subsystem that was changed from our PDR to final design was the medkit delivery system. It is important to note that the actual delivery of the medkit remained the same, we always wanted our medkit to be stored inside the chassis and dropped vertically into the delivery box. The purpose of keeping the medkit within the walls of the chassis was to ensure that in a real-life situation no damage could be done to the package, ensuring the resources would be usable upon delivery. While initially we had designed the storage unit/claw to be directly over the center of the chassis, we shifted the storage location forward so that we would not have to drive directly over the receptacle and could drive just to the front of it. As for the release mechanism of the medkit we had originally designed the delivery system to consist of a claw we would purchase attached to a servo motor that would simply open when Bulkhead was in the correct position. However, in an effort to reduce cost, we pivoted to a cheaper, light-weight alternative, looking to two pieces of cast acrylic that resembled chopsticks. Each "stick" would be attached to a gear that would open when the servo activated. While this design worked to hold the Medkit we were concerned that if the robot experienced any major shifts (down the ramp or over the wall) that the medkit would slide out. It was then that we turned back to a design that more closely resembled the claw idea. This time, 3D printing two halves of the claw and again attaching them each to a gear that would open upon the servo firing. This was the design that we implemented in the final assembly of Bulkhead and used in our demonstration.

5.3.3. Rear Connection Plate

To connect the main chassis to the rear plate, we created a connection component out of $\frac{1}{4}$ inch aluminum found in the shop to reduce cost and ensure stability, as this piece was thicker than both subassemblies it was connecting, the main chassis ($\frac{1}{8}$ inch aluminum) and the rear plate ($\frac{1}{8}$ inch acrylic). Since we changed our design of the rear system from an arm with a wheel

on the end of it to a plate with a wheel on the end of it, we had to redesign the connection between the chassis and rear system as well. To again ensure that this connection plate would form a strong bond between the two subassemblies, we designed the connection plate to cover the maximum optimal surface area of both the chassis floor and the rear plate, as was possible with the material we had available in the shop. With this design specification, as can be seen in Figure 40, we made the primary connection surface area as wide as possible with the material we had access to, in order to prevent bending, and we included eight drill holes on both sides to ensure a strong connection point that would not move. Also, we included the skinnier portion of the piece since room had to be left on that section of the chassis floor for the bolts securing the motors in their mounts. If the bolts for the motor mounts were not located here, then the width of the plate would have been consistent throughout to ensure maximum support to the chassis. After making the piece, we also annealed it, then bent it from 180 degrees to an angle of about 150 degrees so that the bottom of our rear wheel would lie tangent to the same horizontal plane as the bottom of our main two wheels.

5.4. Finite Element Analysis

To ensure that the master assembly, subassemblies, and certain parts of our SaRR remained structurally sound throughout the engineering process, we performed FEA under specific stresses, loads, and dynamic situations, which are detailed below. Along with the full assembly of the SaRR, the subassemblies that we decided to perform FEA on through analysis and concluding where the largest forces occurred were the drivetrain, medkit delivery system (claw), and the rear plate system. The drivetrain was subject to the force from the pulley belts connected to the motor, the medkit delivery system was subject to the weight of the medkit, and the rear plate was subject to the weight of the chassis, normal force with the ground, and friction force with the ground. We elaborate further on the analysis of the individual assemblies in their respective sections below.

5.4.1. Drivetrain

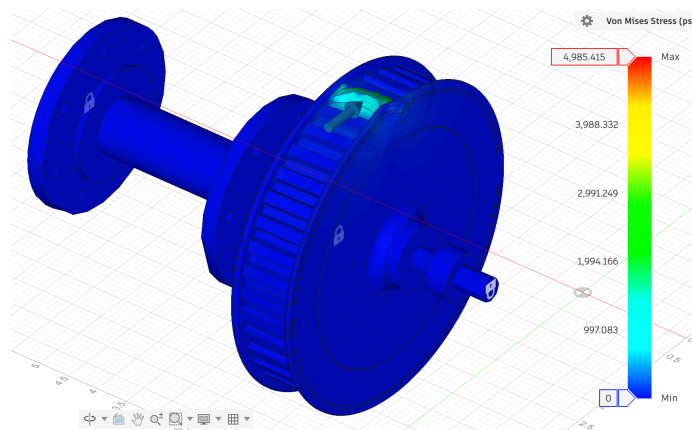


Figure 14: Drivetrain Max Stress (psi)

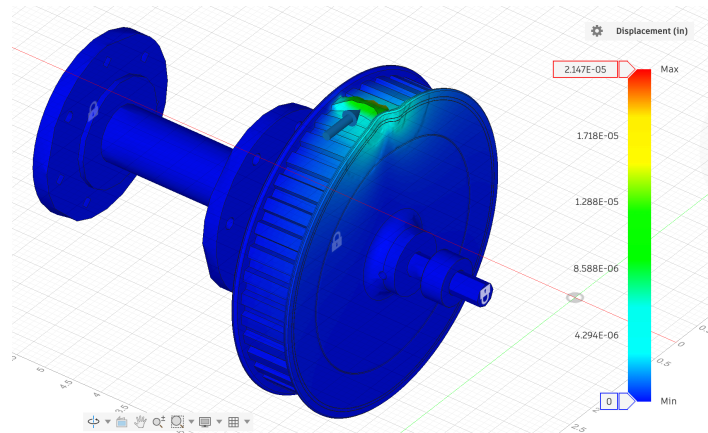


Figure 15: Drivetrain Max Displacement (in)

Figures 14 and 15 above demonstrate simulations that were conducted on the pulleys of Bulkhead's drivetrain. The force was found to be 410 N, which is applied to the pulley as it turns by the belt on the teeth. This is found by dividing the torque of 26 Nm by the radius of the pulley, which was 2.5 inches, or 0.0635 m. With this, you can see that this is the force applied at max power and it demonstrates how there is minimal deformation on the teeth of the pulley at connection point, which allows the team to acknowledge that the teeth will endure the continuous load that is applied tangentially by the belt.

5.4.2. Medkit Delivery System (Claw)

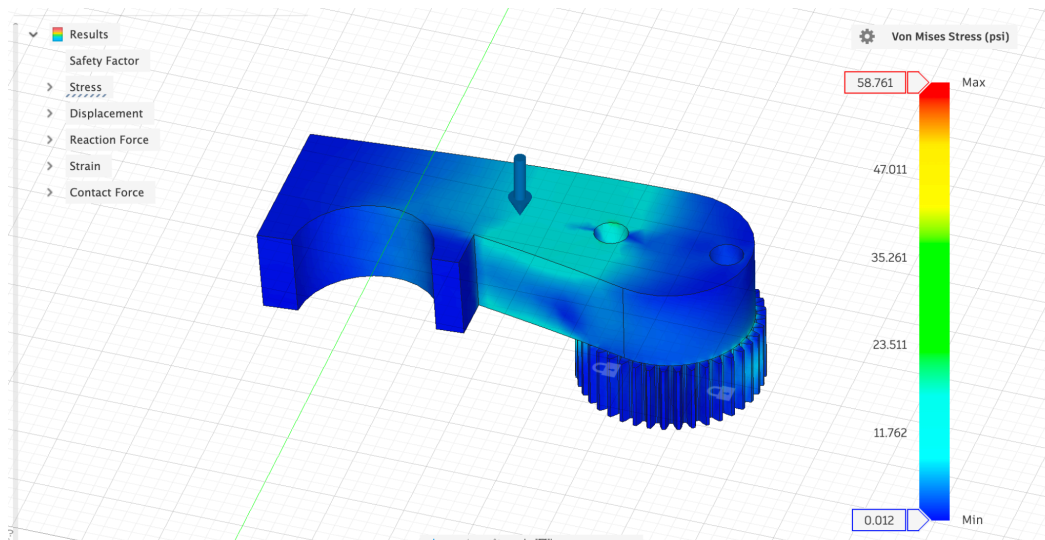


Figure 16: Max Stress On Claw With 0.5 kg Applied (psi)

The main thing we were looking for when performing the FEA on the medkit claws was to make sure that the robot could drive properly without snapping or bending so far from neutral

that delivery of the medkit would be impossible. As we can see in Figure 16 above, the maximum Von Mises stress occurred over the area between the grip and the gear mount, with a max value of about 11.762 psi. This value is very low and as we can see from our demonstration must be much lower than the allowable shear stress for this part.

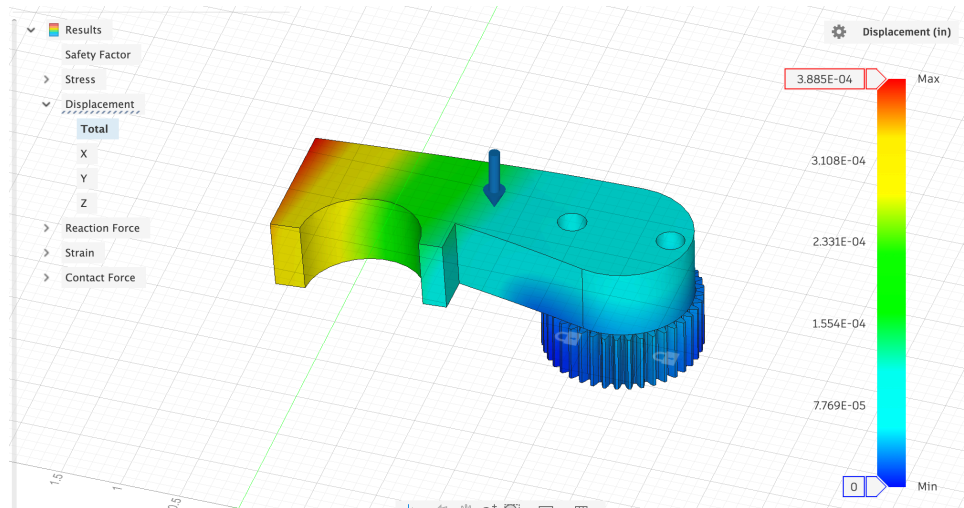


Figure 17: Max Displacement of Claw (0.5 kg applied downwards)

Again, we can see in the picture above the maximum displacement in inches of the claw when half of a kilogram is applied in the downward direction. While we technically didn't have a max displacement value to stay under, we really just wanted to minimize the bending as much as possible so that nothing was messed up during the actual delivery of the medkit. Taking the max displacement to be about 3.88×10^{-4} inches, we see that our claw would barely bend at the end, meaning we could expect the claw to remain relatively parallel with the top of the chassis while driving.

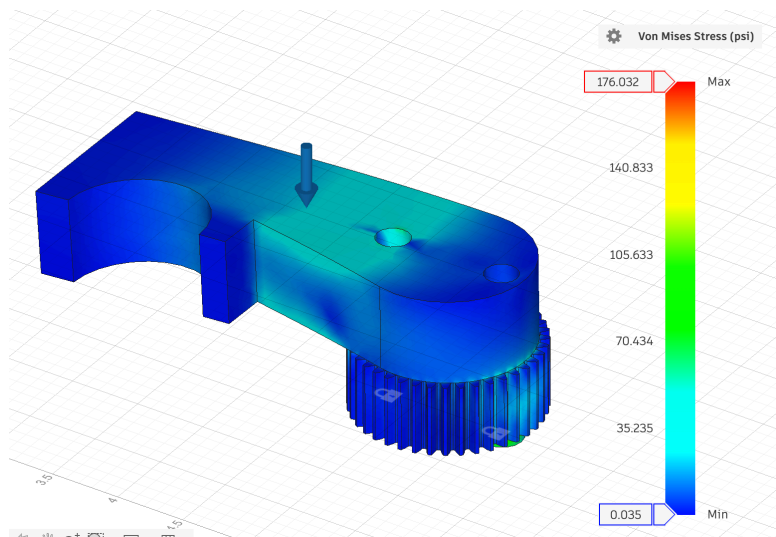


Figure 18: Max Stress On Claw (psi) (1.5 kg Applied Downwards)

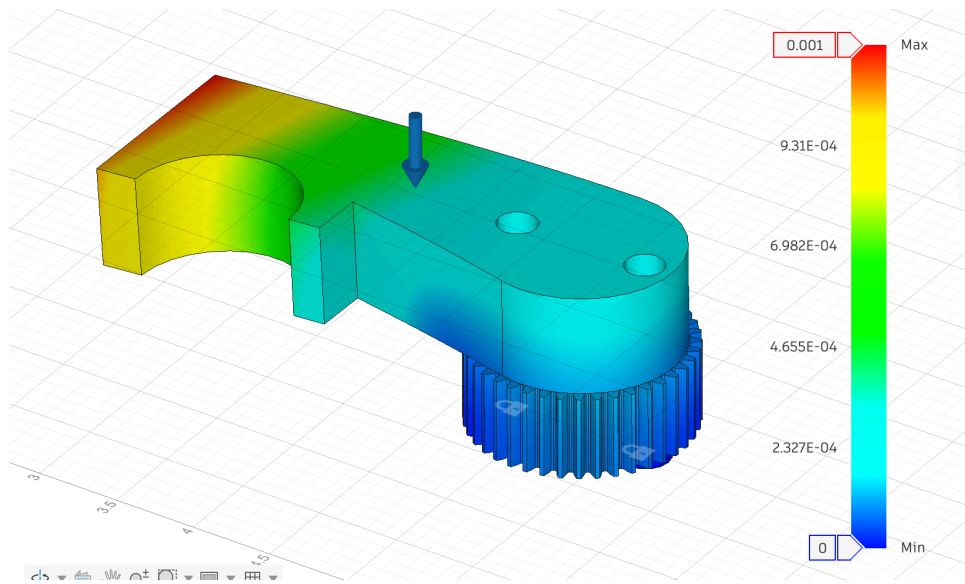


Figure 19: Max Displacement of Claw (in)(1.5 kg Applied Downwards)

The above two pictures show stress and displacement on the claw for the scenario of the SaRR dropping over the wall. Due to the SaRR dropping from a height of a foot off the ground, we anticipate this simulation to be similar to the stress and displacement experienced during the drop test, which we expect to be no more than three times the weight of the medkit, so the load experienced by one side of the medkit delivery system (one claw) was approximately 1.5 kg (3.3 lbf). The stress simulation shows that the entire claw system experiences relatively low levels of stress, with a maximum value of 176.032 psi. A concentration of stress between the gear and grip mount is observed, very similar to the analysis for a 0.5 kg weight. This deviation is again related to the balance between the weight of the Medkit and its mount to the chassis, which is intended to prevent the claw from moving during acceleration loads, thereby keeping the Medkit secure. However, the stress magnitude is very low and within the range of maximum allowable shear stress. In addition, the displacement simulation displays a maximum value of about 0.001 inches in a small concentrated area on the left edge of the system since it is not attached to anything. The rest of the claw barely bends, so the claw is expected to remain relatively stable and parallel to the top of the chassis throughout the course.

5.4.3. Rear Plate

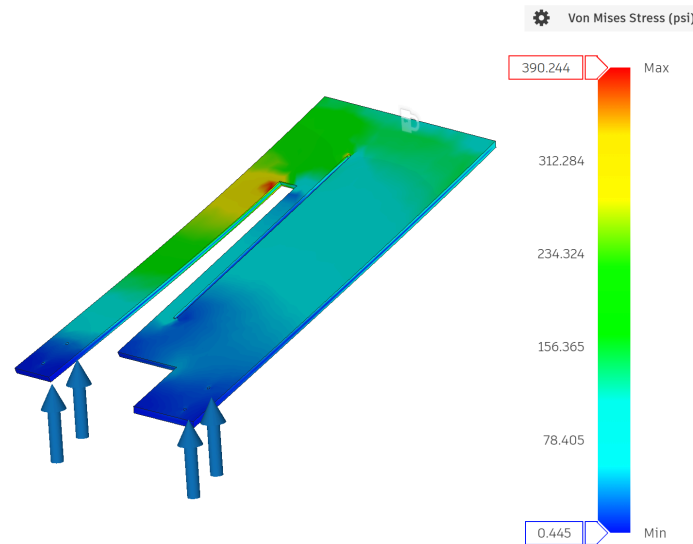


Figure 20: Max Stress On the Rear Plate (5 lb Applied Force)

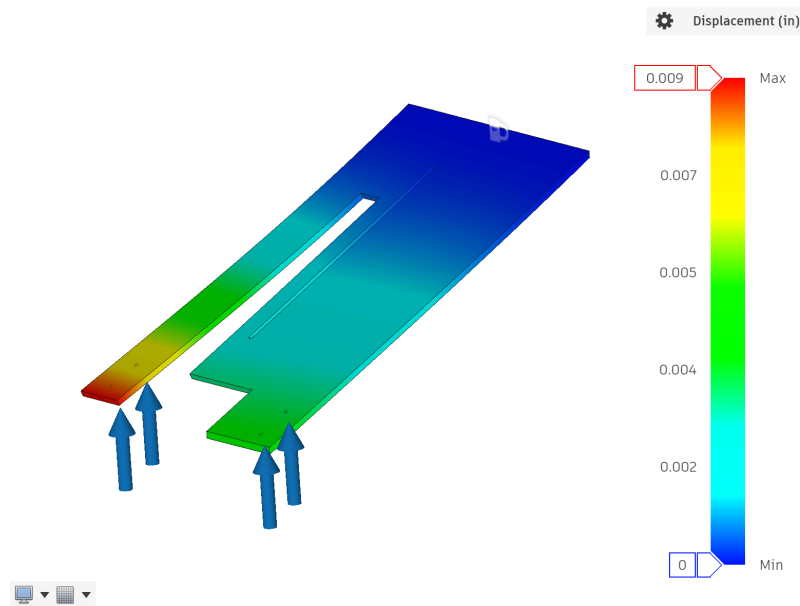


Figure 21: Max Displacement On The Rear Plate (5 lb Applied Force)

Above in Figure 20 and Figure 21, you can see simulations of the rear plate undergoing an applied force from the reaction of the plate coming in contact with the ground after a drop test. This simulation applies to the drop test that is conducted before the final demonstration, but this simulation can also be considered when analyzing the stress and displacement of the plate after Bulkhead goes over the wall. As seen in Figure 21, the bending occurs the most in the middle of the plate, which is realistic because there was an aluminum support at each side, one to

hold the axle for the wheel and another to connect it to the chassis. It is only natural that the stress would occur in the center of the plate between these two pivot points. Our team countered this in the physical model by adding two aluminum 6061 support plates underneath the middle of the rear plate on the left and the right in order to counter this bending. This proved to be useful as we observed there to be no plastic deformation in the rear plate during the drop test.

5.5. Control System

Figure 22 below shows the wiring diagram for Bulkhead's electronics, including all sensors and actuators. Much of the electronics system was kept consistent with the sample robot built earlier in the semester for MAE322. The battery is connected to a light switch, which is then connected to a fuse box, which then distributes power to the motor controllers for the drivetrain. The light switch is an easy way to positively cut power to the entire system, and the fuses in the fuse box prevent current draws of over 30A per motor. Additionally, the battery and motor controllers are both grounded via ring terminals to a grounding box to complete the power circuit. The motor controllers are connected to the Teensy 4.1 Microcontroller, which commands the motors to spin at speeds based on the sensor inputs. Lastly, there is a servo motor that does not require a motor controller and is connected directly to the Teensy.

Bulkhead takes in inputs from six sensors: two side infrared distance sensors, 1 front infrared distance sensor, two photoresistor light sensors, and one inertial measurement unit (IMU). The IMU's primary purpose is for state switching. To switch from a wall-breaching state to a light-finding state to find the medkit basket, the IMU looks for a spike in vertical acceleration exceeding 1.5g. We also found that when climbing the ramp, the x-acceleration (which is normally $< 2 \text{ m/s}^2$ in normal driving operations), increases/decreases to $\pm 2\text{-}5 \text{ m/s}^2$ when ascending/descending the ramp due to the IMU being tilted. Therefore, the x-acceleration was used to switch between ramp ascending and descending states. The front distance sensor was used in a similar way to determine when Bulkhead had arrived at the wall or the medkit delivery basket. In contrast, the right and left distance and light sensors made the most significant contributions to Bulkhead's control algorithm.

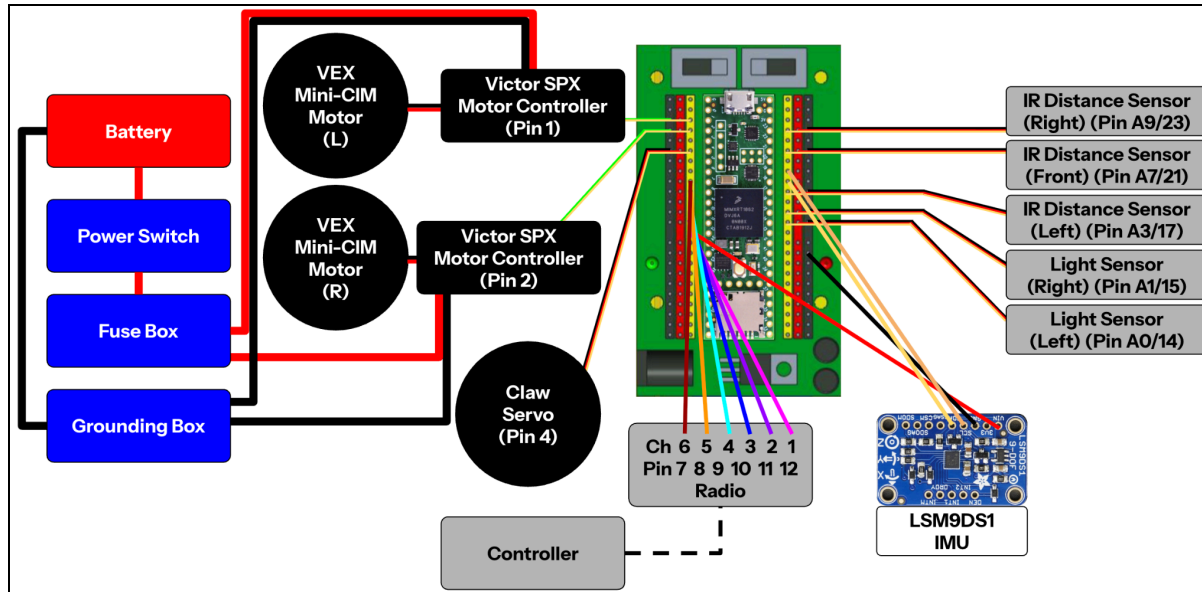


Figure 22: Bulkhead Electrical Diagram

Figure 23 below shows the followLight function that the team created to connect sensor input from the light sensors to outputs to the motors. Although not the simplest function that could be written, the function is quite robust and clever because it accounts for the fact that photoresistors can vary in their sensitivity. For example, the right and left sensors could return different values for the same brightness of light, which would pose problems for driving straight. The followLight function eliminates this problem by defining a maximum value that the sensor returns. Then, a scaled measurement of the light relative to the sensor's maximum is calculated. Now that the measurements are normalized and on the same scale, the values of the two sensors are summed to define the total light seen. Since the total light is known, the values on the left and right sensors can be compared as a fraction of the total amount of light seen to determine if the light is brighter on the left or right. Then, using proportional control, the drivetrain motors are commanded to spin at a speed that corresponds to the proportion of light seen on each side of the robot. That is, if the light is brighter on the left, the left motor spins slower and the right motor spins faster to turn Bulkhead to the left and center the light.

```

402 void followLight() {
403     // Motor range from 1000 to 2000. 1500 is stop. 1000 and 200 are full forward or reverse (different on left and right)
404     // Need to take in leftIRVal and rightIRVal and convert those to speeds
405     // Bright Light = 200?
406
407     readSensors();
408
409     // Manually calibrated max values
410     // right ranges from 750 (dim) to 250 (bright)
411     // left ranges from 250 (dim) to 50 (bright)
412     double maxRight = 1000;
413     double maxLeft = 1000;
414
415     // Scaling because the right and left are drastically different
416     double scaledRight = rightIRVal/maxRight;
417     double scaledLeft = leftIRVal/maxLeft;
418
419     double totalLight = scaledLeft + scaledRight;
420
421     // Determining which side sees a greater fraction of the total scaled light. Drive toward that side
422     double fracRight = scaledRight/totalLight;
423     double fracLeft = scaledLeft/totalLight;
424
425     // Use the following lines to print the sensor values to serial monitor for debugging
426     Serial.println("Left: " + (String)(constrain(1500 - 500*fracLeft, 1000, 1500)));
427     Serial.println("Right: " + (String)(constrain(1500 + 500*fracRight, 1500, 2000)));
428
429     L_Servo.writeMicroseconds(constrain(1500 - 500*fracLeft, 1000, 1500)); // sets the servo position
430     R_Servo.writeMicroseconds(constrain(1500 + 500*fracRight, 1500, 2000)); // sets the servo position
431
432     // Best practice to delay
433     delay(100);
434 }

```

Figure 23: followLight Function for Proportional Closed-Loop Control

Figure 24 below outlines the control algorithm for switching between states in autonomous operation. The diagram shows the different states that Bulkhead switches between, as well as the criteria for switching between the states. The state switching occurs based on an integer variable “state” which starts at 0 and is increased by 1 every time Bulkhead completes a state. The autonomous code consists of a series of if-statements that check the value of the state counter variable, making it very straightforward to switch between states because as soon as one state is completed, the state counter increases. Therefore, it is impossible for Bulkhead to get confused and go backwards into a previous state or skip states. Every state’s exit criteria must be satisfied for the next state to start. A video of Bulkhead running through all the states in the code can be found in Appendix B, and the final code is in Appendix C.

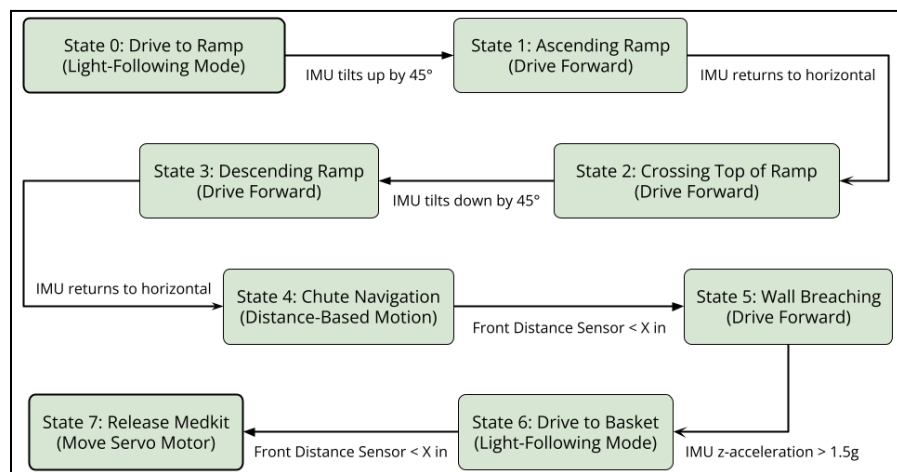


Figure 24: Outline of Control Algorithm for Autonomous Operation

6. Test Results

Prior to the final demonstration, each part of the course was tested for the robot's ability to autonomously achieve individual milestones. When Bulkhead was able to successfully navigate the ramp, it gained too much speed while descending and crashed into the chute. As mentioned in 5.2.2, at this point, gravity is working with our timing belt, and it hurts us. It was difficult to keep the speed slow enough to maintain control, especially with little room for error horizontally. Instead of easing over and taking a gradual descent, the normal acceleration continued, and the SARR did not have time to adjust. Ultimately, the right tire slipped off the ramp, and it caused the light sensors attached by a standoff in the front of the SARR to break, slowing down the overall testing process. This happened a few times, and after the third, we just started duct taping it over so that we could get quicker iterations and start testing other mechanisms and designs. 3D printing a new set for the final attempt.

Another major setback was the set screw dilemma. We had initially incorporated a set screw into a D-Shaft that would hold the 3D printed pulley to the shaft. The pulley was spun up by the belt, and that would move the shaft, all of that through the singular set screw. This issue was pressing and major. It took us a while to move off the PLA printing and into a metal component. In hindsight, after the first shear, we should have machined a separate piece out of metal to attach to the pulley, and run 2 set screws through that metal piece. In this way, the force would be distributed over 2 set screw areas, and keeping the attachment point metal, to ensure that nothing shears. This was the solution, but it took us costly iterations during important testing time to get to this point. After that, we could start testing different features of our run-through.

After conducting various iterations of the wall-traversal, the team observed that the motors stalled after reaching the wall, and the belt was losing tension. We observed this was due to the motor sliding in its placement, and allowing slack in the belt. This slack caused the belt to slip continuously when attempting to get over the wall. There was not enough constant tension to allow the tires to get spinning. To mitigate this, we created plates to attach to the motor mounts on the bottom side of the chassis, in order to increase the surface area of contact for the connection (think of it as a massive washer). This helped, and with more time, we would implement a belt tensioner that would keep the tension high and strong throughout the process.

Finally, while testing for movement to the Medkit basket and Medkit delivery, Bulkhead successfully navigated towards the light on the basket and dropped the Medkit after a duration of about 30 seconds, which was approximately 30 feet from the wall. The max speed reached by the robot was 1.047 ft/sec. The medkit was held by the medkit delivery system and was dropped off once the SaRR drove over the basket. The code instructed the servo to rotate once the SaRR had driven far enough over the basket, and rotating the servo rotated the claw gear mechanism outward, thus releasing the claw and dropping it into the basket.

During final demonstrations, the team began with a closed-loop testing of the SaRR. During previous autonomous testing, there was no state for Bulkhead to complete a 180° turn while facing away from the light initially. Therefore, the first task of the day was to add a state before State 0 (State -1) to manually turn Bulkhead. This State -1 was timer-based, just to turn Bulkhead around so that it was vaguely pointed in the right direction. Then, we realized that for some reason, the code logic was backwards, so that when the light drifted toward the right of Bulkhead, Bulkhead would turn left, away from the light. To fix this, Bulkhead was jacked up so that the wheels could spin without the robot going anywhere, and we iterated the code to reverse the logic.

Continually, during the closed-loop testing, since the SaRR spun in a circle repeatedly, unable to find the light source of the ramp, this caused the medkit to fall out of the medkit delivery system. The medkit did not fall out of its delivery system during any of our testing before the demo; however, we only had time to test the medkit delivery from after the ramp to the basket by following the light, which involves driving in a straight line mostly and not much rotation. Because of this, we didn't test the stability of the medkit in its delivery system for the part of the course before the wall, so we were unable to account for the possibility of the medkit falling out when the SaRR rotates to find the light in the ramp, which occurred during our demo and caused our medkit to fall out.

Next, we decided to attempt the ramp in an open-loop setting to see the navigation and maneuverability of the SARR. When traversing the ramp, as we entered the descending portion of the ramp, we accelerated considerably, causing the back wheel to lose touch with the ground and decrease steering/maneuverability. This resulted in us hitting the rightmost side of the chute, and Bulkhead twisted sideways and snapped the right distance sensor and light sensor. During this crash as well, the screws that were holding the left drive motor were ripped out of the motor, allowing the motor to move within the chassis and disconnecting the motor pulley from the timing belt. Because of this, one of the drivetrain systems no longer worked, so we could no longer drive our SaRR since fixing this issue would have taken longer than the time remaining to perform demos. Since this issue occurred while performing an open-loop traversal of the ramp, we unfortunately could not attempt an open-loop traversal of the other sections of the course. Since our closed-loop trial was not successful, we aimed to successfully drive through the course with an open-loop traversal; however, we were unable to attempt the chute, wall traversal, and medkit delivery after the failure of one of our drivetrains. If this failure had not occurred, or if we had more time to fix the issue, then we would have attempted an open-loop traversal of the entire course.

Finally, as stated before, there was an issue with securing the Medkit due to rotational movement and transient shock loads experienced during the course. Since the claw is designed with a small opening, continuous vibration while navigating towards the ramp caused the Medkit to become dislodged from the claw and fall out. It was previously assumed that the size of the opening and the magnitude of shock loads would not cause a severe issue, but further testing

with the Medkit proved otherwise. Ultimately, final test results for wall-breaching and Medkit deployment speed and duration could not be obtained. If there were to be a future iteration of Bulkhead, the chassis should be designed to be narrower so that there is more leftover space on the ramp for maneuverability, and the robot is less likely to fall off the ramp and break. In addition, more time for calibration and focus on light detection would have fortified Bulkhead's directionality, which had direct implications for State 0 of our demonstration, but also indirect effects of later states by causing the medkit to be deployed, and making the ramp ascent unsteady.

7. Conclusion

Bulkhead is a robot designed around being lightweight, simple, and commercially sellable. The design incorporates two large wheels and a bottom-heavy chassis that is freely rotating so that Bulkhead always remains upright, even during the wall traversal. All components are housed within a central sturdy chassis, protecting itself from the challenging environments it finds itself in. The Medkit sits in the center of the chassis, and instead of an arm delivery system, it simply falls out of a hole in the bottom of the chassis. To navigate autonomously, Bulkhead uses a frontal proximity sensor and IMU to determine when to change states, two light sensors for light navigation, and two side proximity sensors for chute traversal. This design philosophy keeps in mind the real-world scenarios where Bulkhead can be deployed and manufactured in bulk to emergency scenarios with ease. From testing, areas of improvement were noted and will increase the chances of successful rescue operations.

Further work on Bulkhead should consist of a detailed analysis of how much weight can be removed from the structure by introducing cutouts and/or pockets in the chassis. This would further fulfill Bulkhead's goal of being a lightweight, robust, and easily producible/reproducible SaRR. Furthermore, more iterative design work involving the correct tensioning of drive belts, security of the light sensors, and stability of the claw is required to increase the reliability of the drivetrain and autonomous navigation.

While Bulkhead demonstrated many promising capabilities, the team also faced several challenges – particularly time constraints and unforeseen risks, which limited the team's ability to fully test and optimize Bulkhead. A two-week computer downtime, manufacturing delays as a result of said downtime, and last-minute integration difficulties condensed our desired testing phase and prevented the robot from achieving its full potential during its demonstration. Despite these challenges, the team's collective dedication, late nights in the lab, and willingness to problem-solve under pressure were instrumental in producing a fully functional prototype that embodied the project's core spirit.

In conclusion, Bulkhead represents a strong foundation for a deployable search and rescue robot, and with additional refinement and extended testing time, its simple yet robust design could realistically evolve into an operational system capable of performing in real-world rescue missions. Despite the setbacks faced throughout this process, our team takes pride in the

immense effort, perseverance, and collaboration that brought Bulkhead from just a concept to a reality. With future work, Bulkhead could one day find its way to saving lives out in the field.

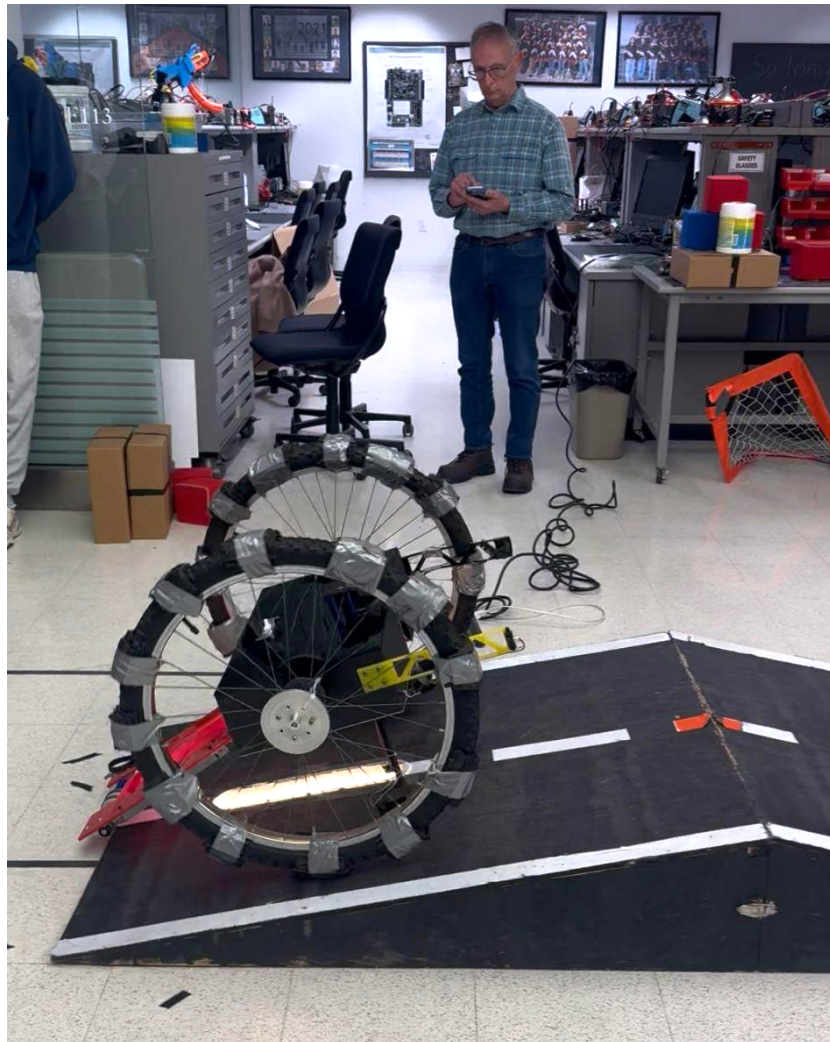


Figure 25: Final Demonstration

8. Appendix

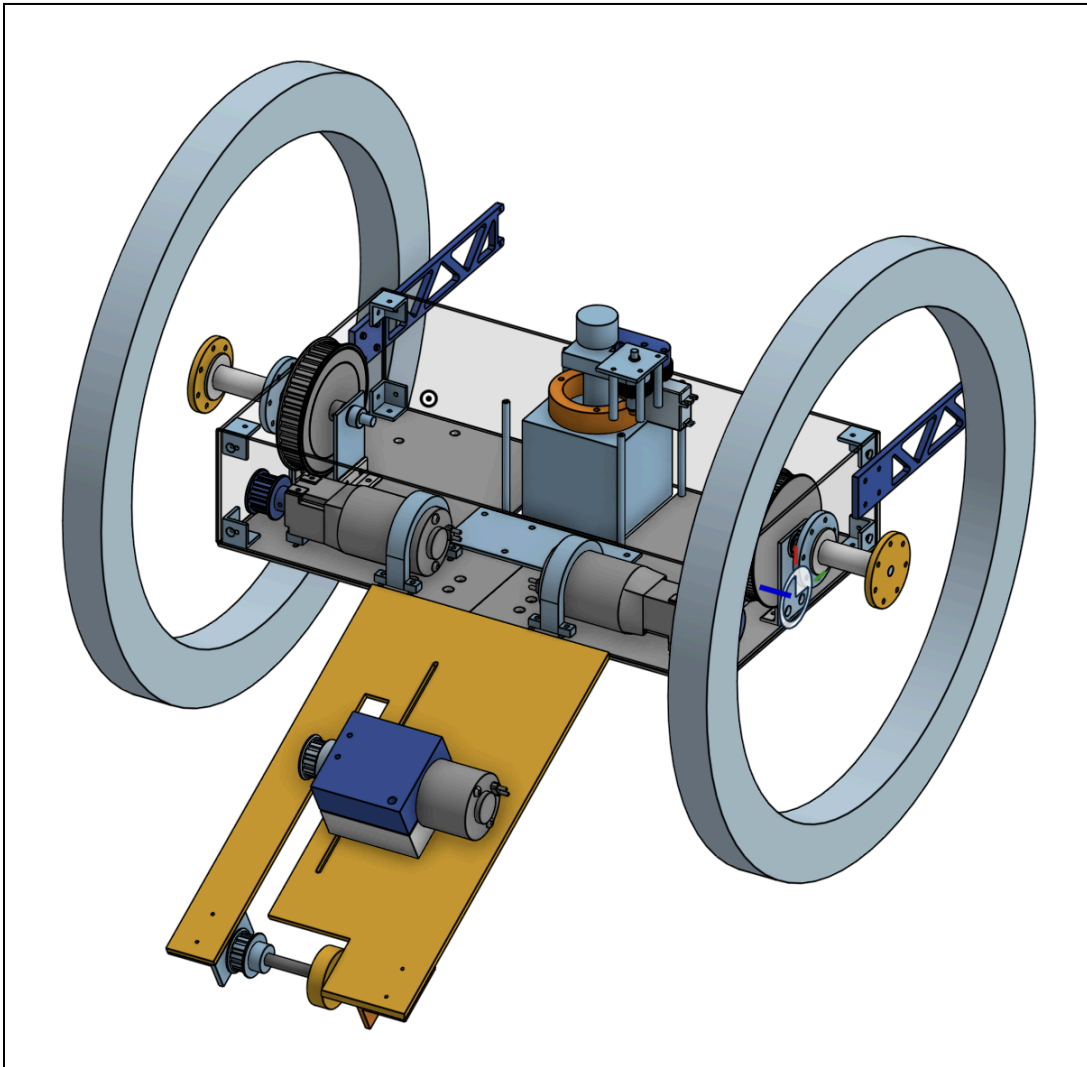


Figure 26: Full SaRR Assembly

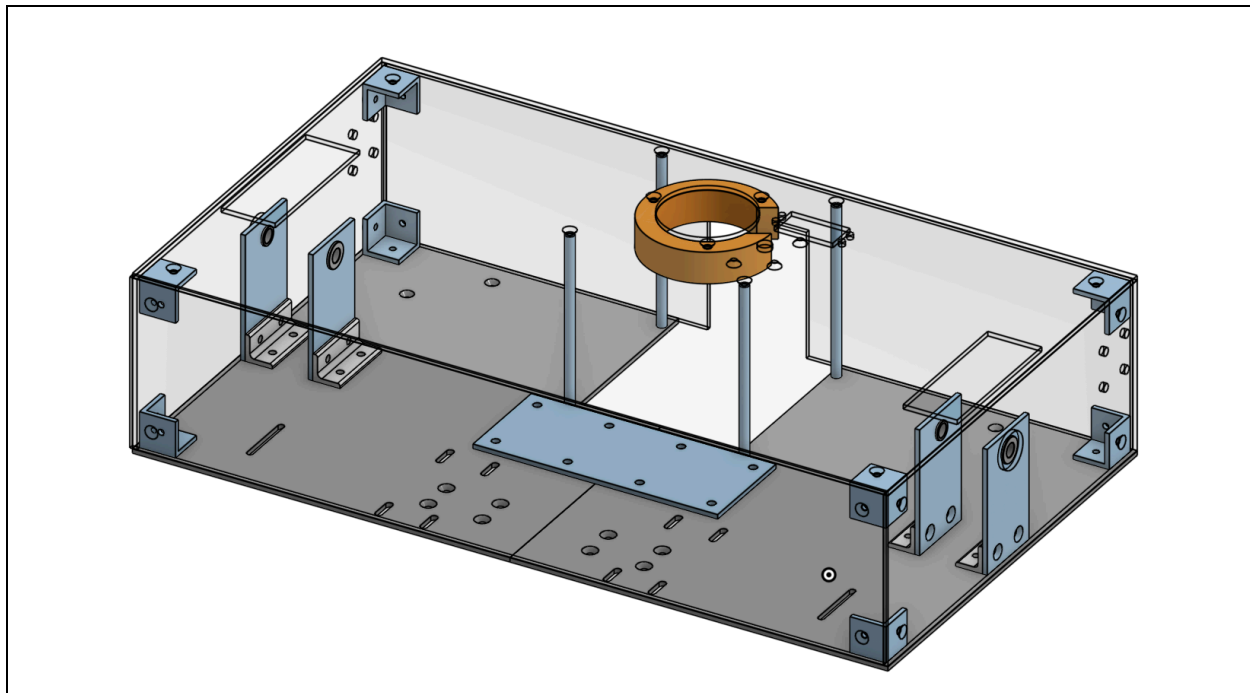
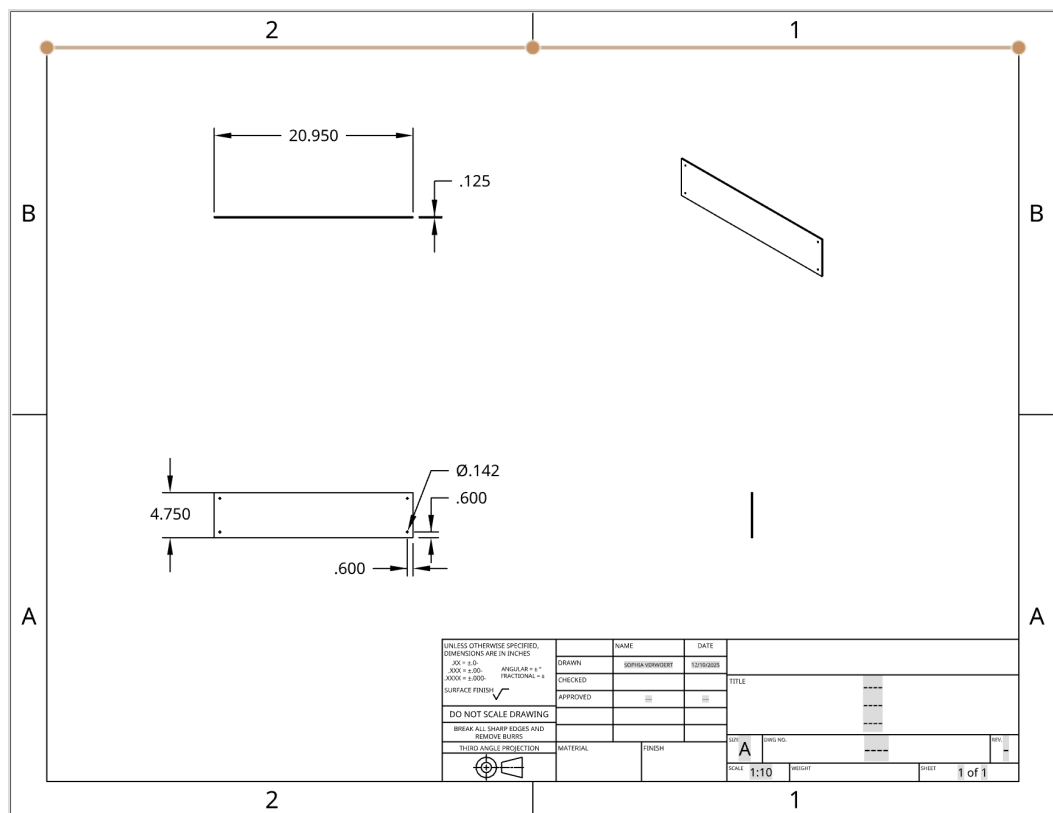


Figure 27: Chassis Assembly



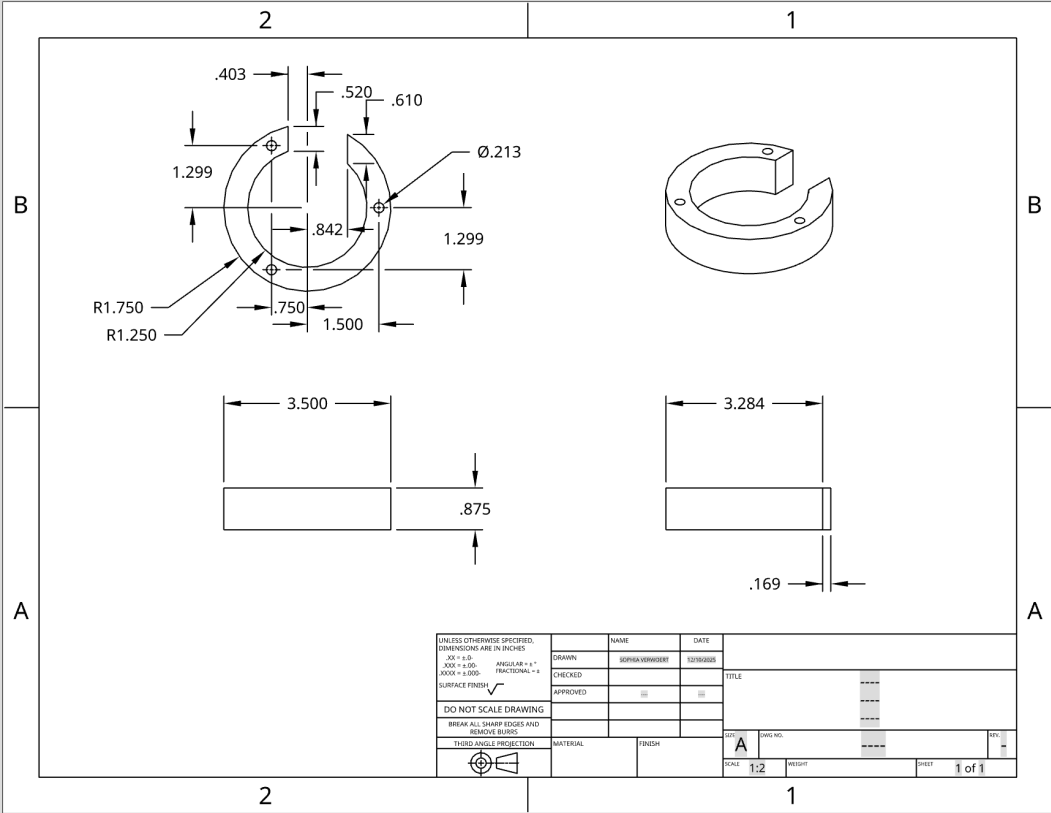


Figure 29: Medkit Claw Hold

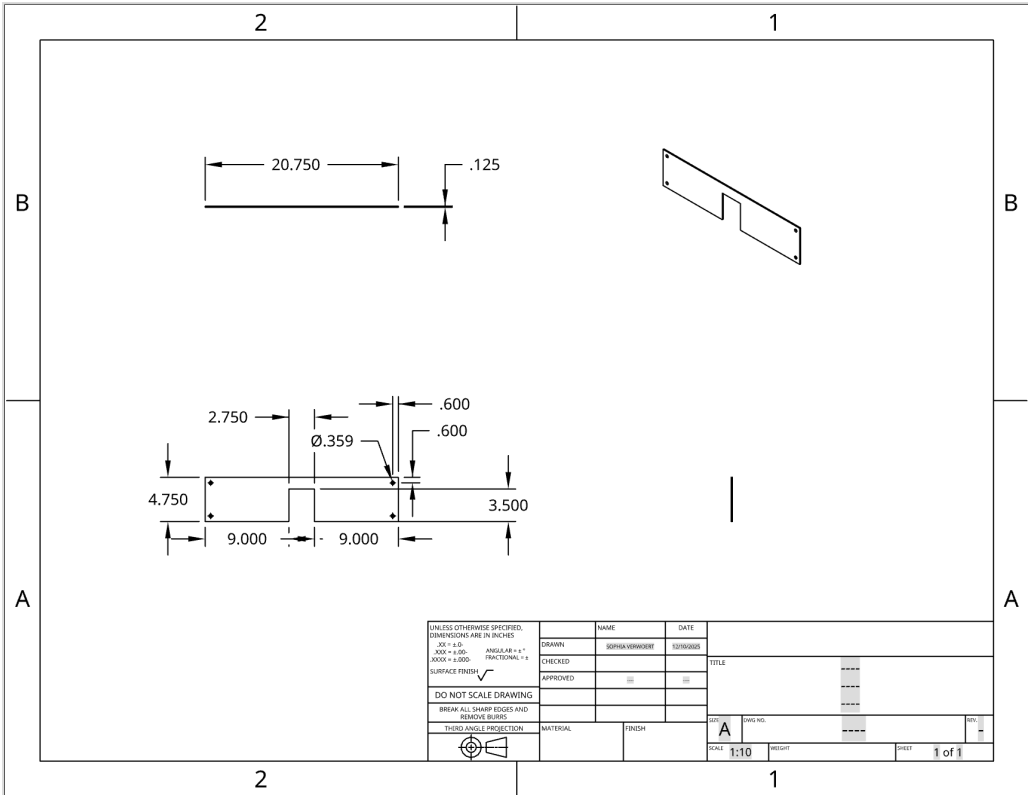


Figure 30: Front Panel

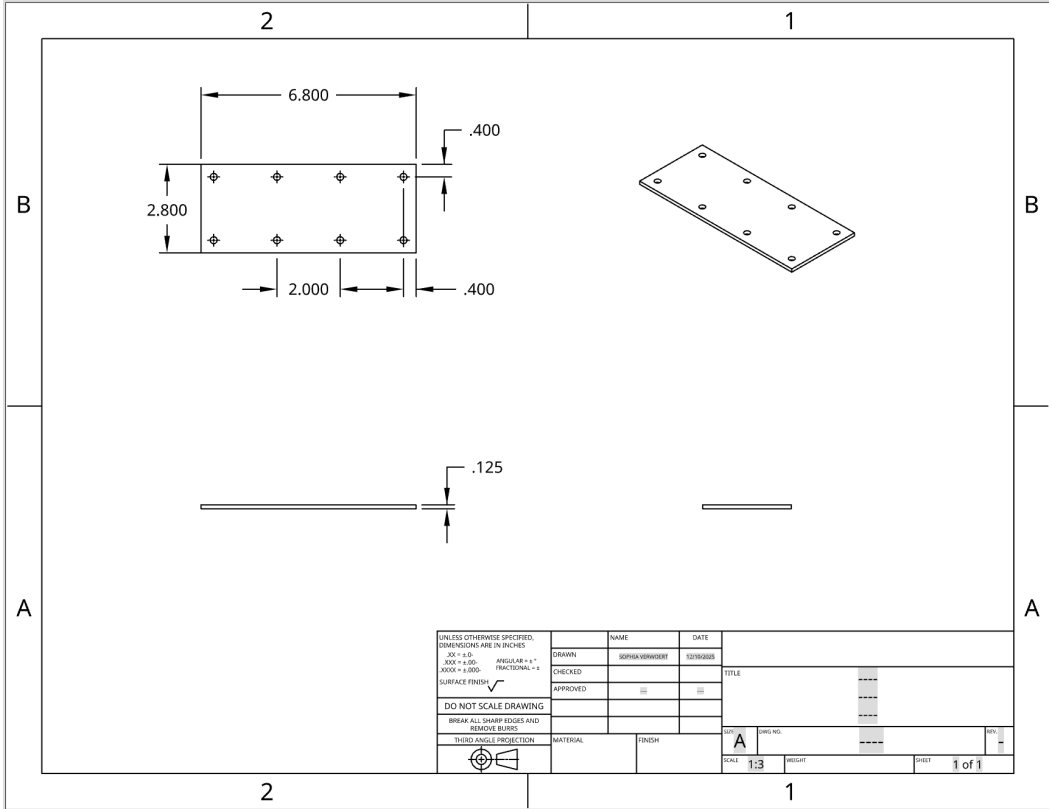


Figure 31: Bottom Panel Connector

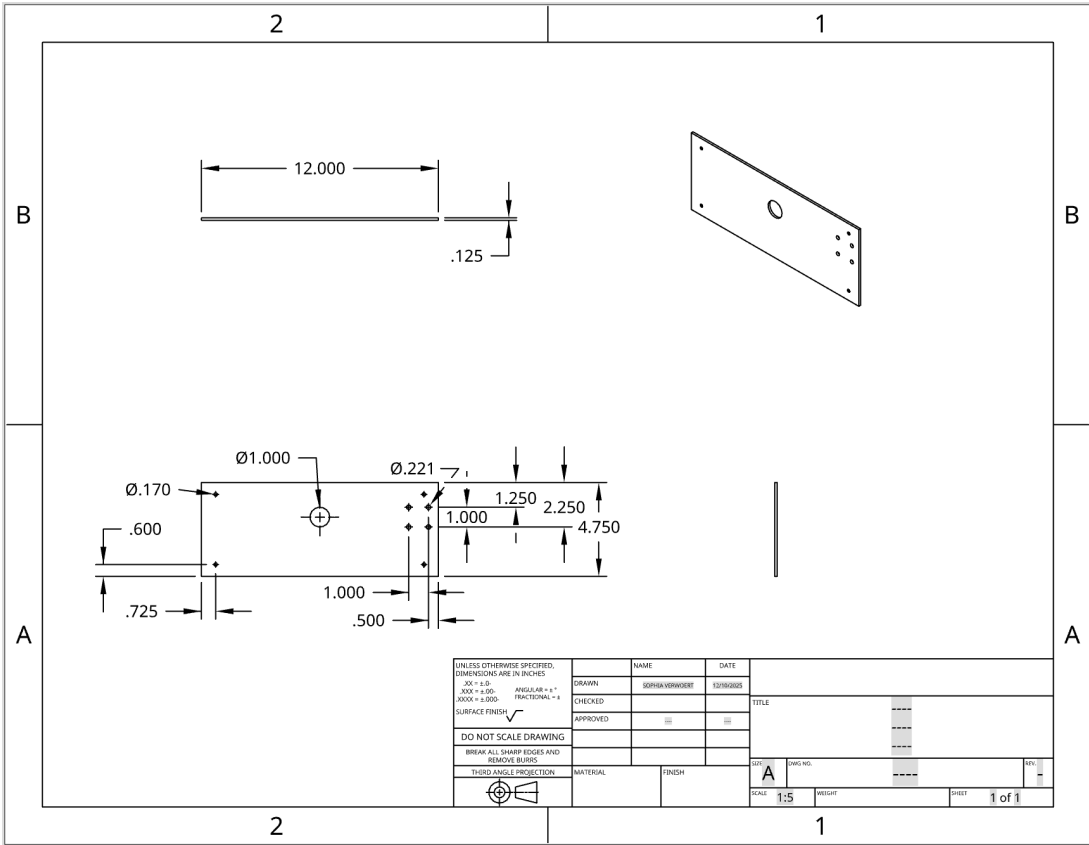


Figure 32: Side Panel (x2)

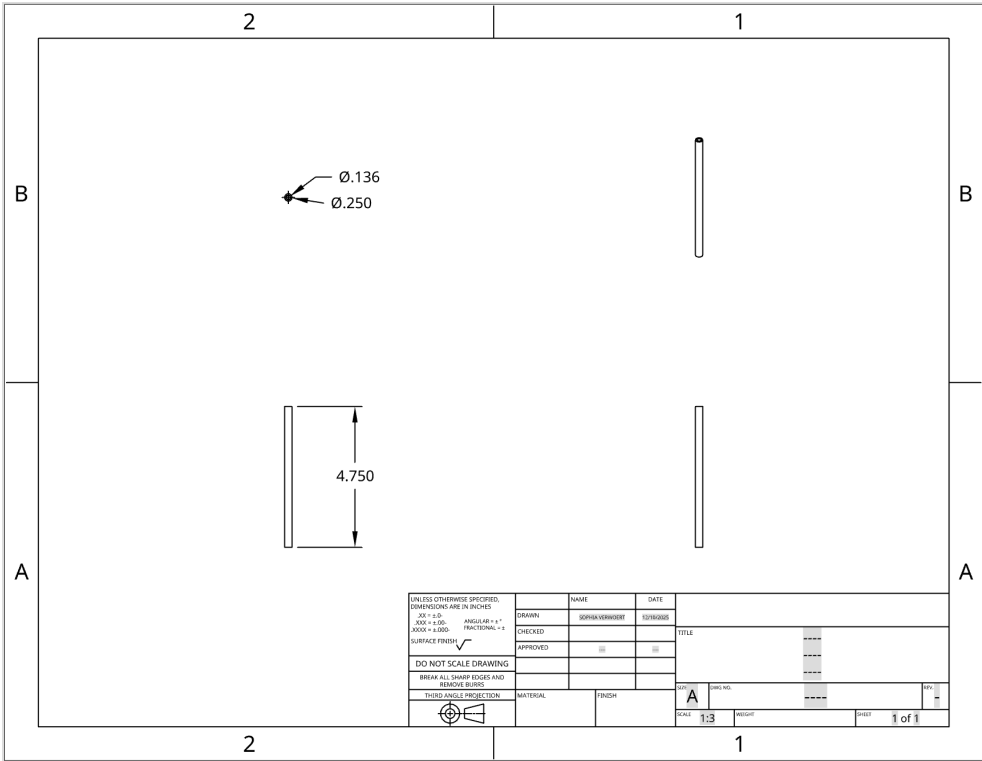


Figure 33: Medkit Hold Standoff

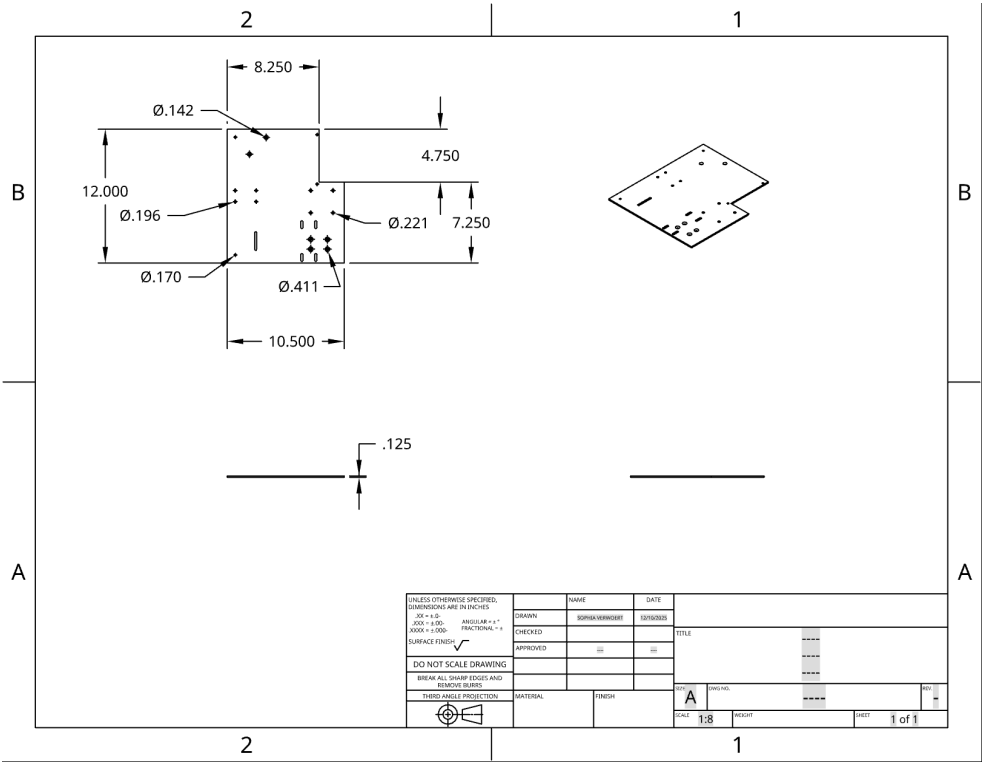


Figure 34: Right Back Panel

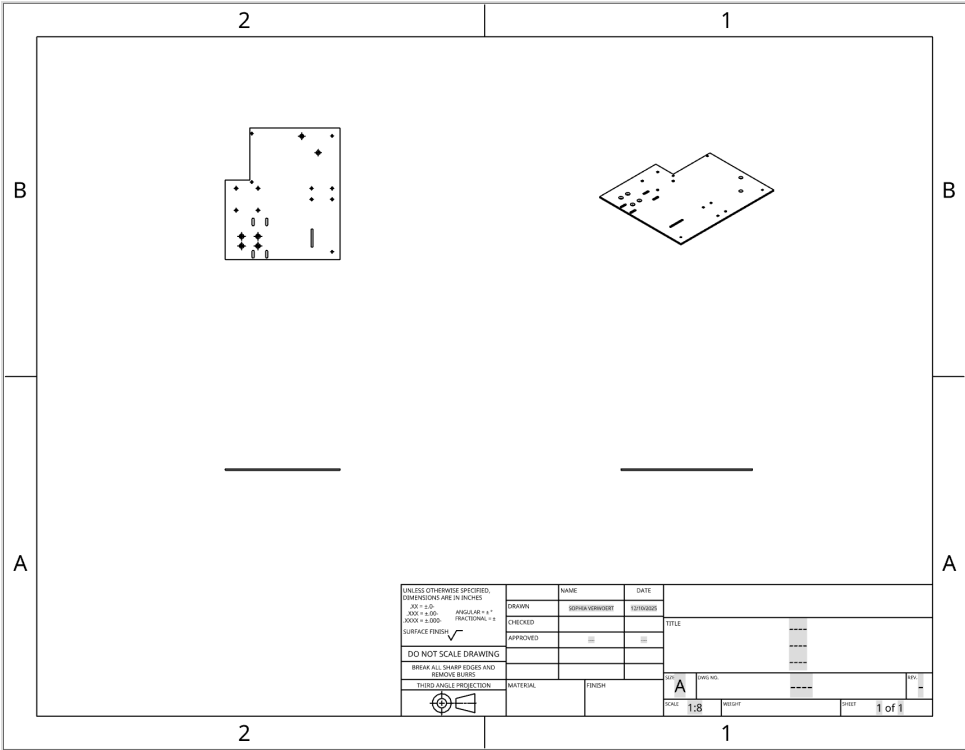


Figure 35: Left Back Panel

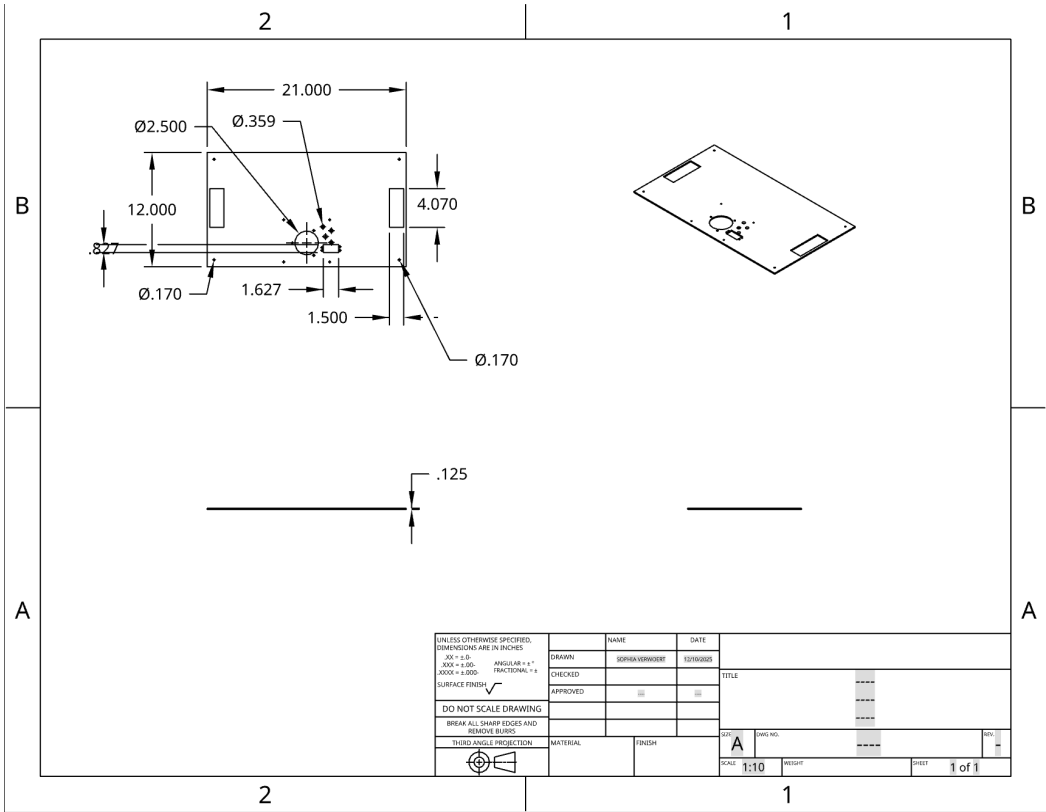


Figure 36: Top Panel

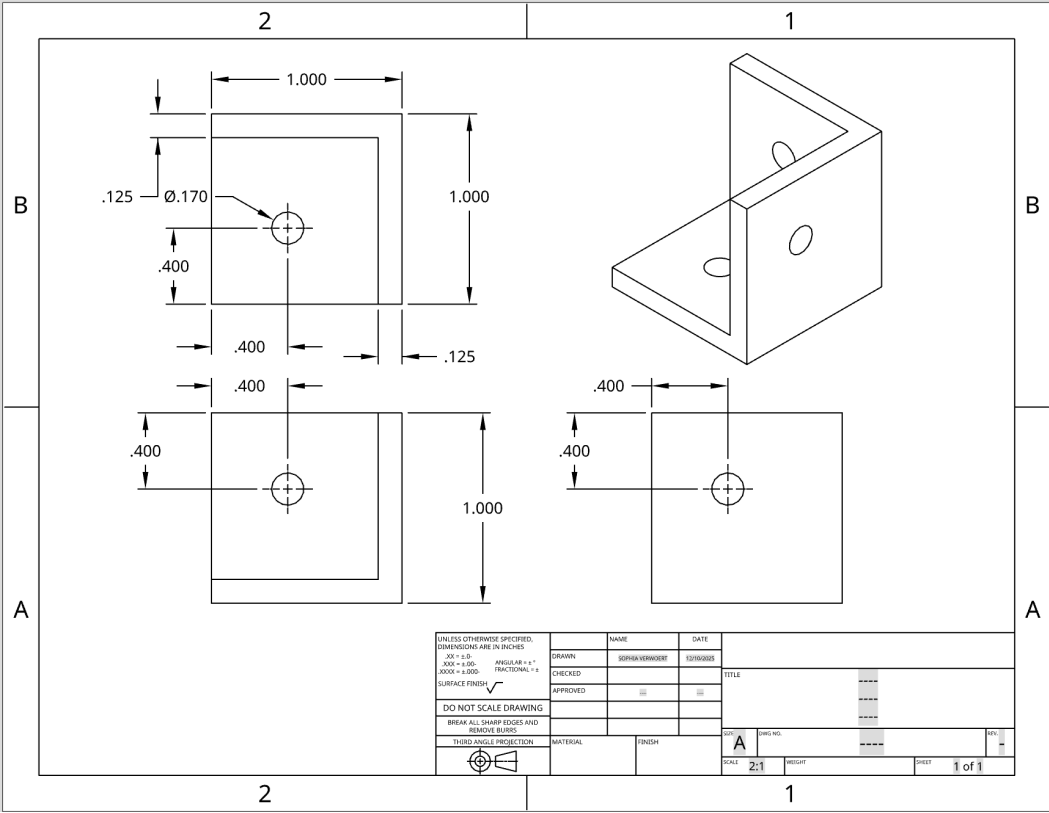


Figure 37: Corner Brackets

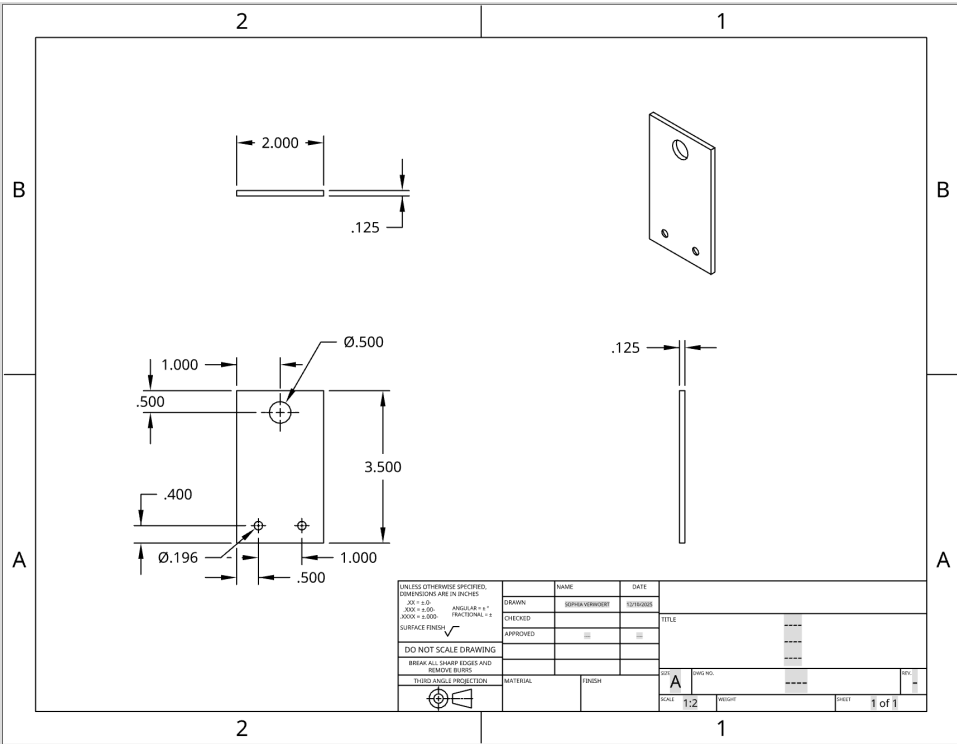


Figure 38: Axle Mount

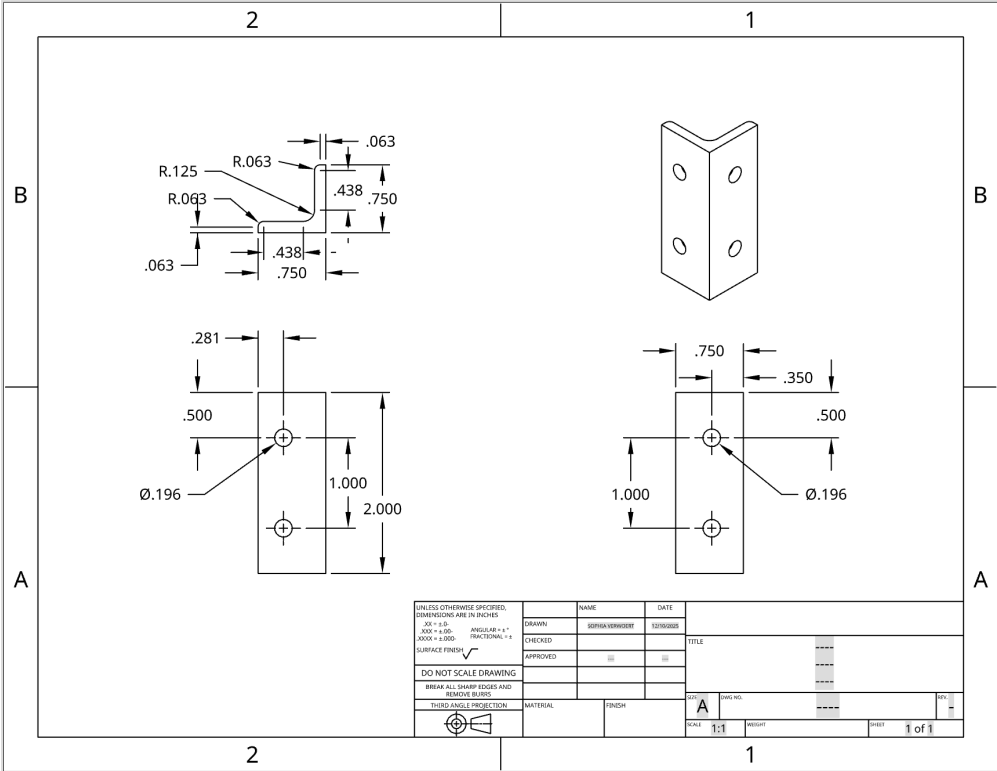


Figure 39: Chassis Axle Mount Support

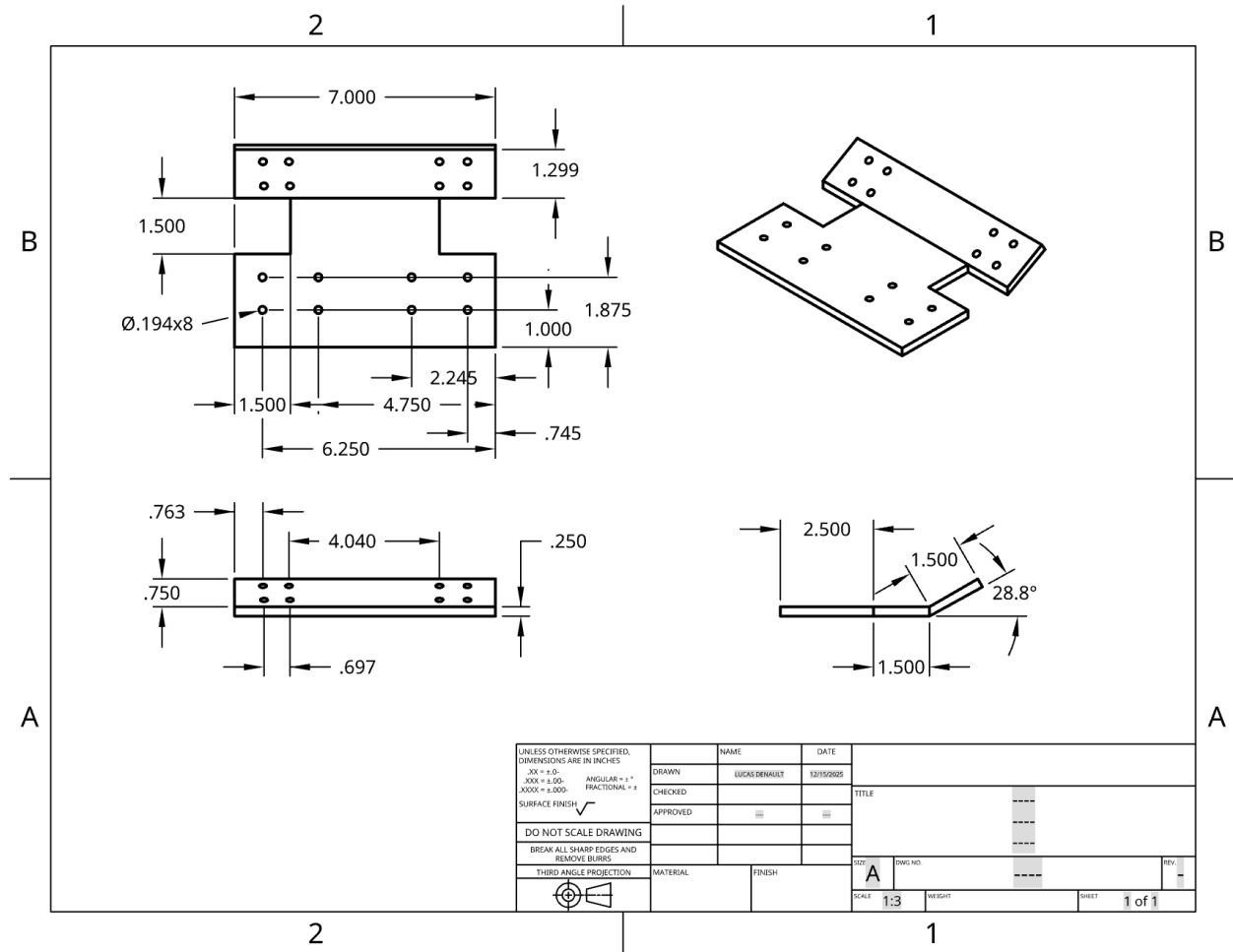


Figure 40: Rear Platform to Chassis Connection

[A] source for acrylic properties

<https://www.interstateplastics.com/materialspecs/cast-acrylic-product-info-062718.pdf?srsId=AfmBOooz4HkdwhSv7a34kmZMkBLITr6ualQWZtCxeaHmlDhLZlBe3WvE>

[B] Video of Autonomous State Switching:

https://drive.google.com/file/d/16NGjGSmLg5ON-P7Ppx_9r2qcThJWIV19/view?usp=drivesdk

[C] Full Code:

<https://drive.google.com/file/d/1JKrj9G5BDe3kfAEmUY7zrvBW3KN5v0mW/view?usp=sharing>